Concrete Lebesgue and Hausdorff Integral Applications

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1 Concrete Lebesgue and Hausdorff integral applications

A first collection of **16 fully worked examples**, written for an analysis student. Start with the problem index, then open the separate solution that you want. The examples favor explicit

computations, stated hypotheses, and direct arguments explaining why the integration theorem applies.

- Measures and theorem conventions: normalized Hausdorff measure, completed-product sections, signed integrability, area and coarea.
- Problem index: no computed answers.
- Counterexample atlas: precise failures and sufficient conditions for exchanging sums, limits, derivatives, and integrals.
- Full worked collection, PDF and editable LaTeX.
- Compact guide, PDF and LaTeX.
- Source references, source snapshot hashes, and verification ledger.
- Machine-readable index without results, and full metadata with hypotheses, results, and review status.

1.1 Find a technique again

From the repository root, using Python 3 with its standard library:

```
python3 integral-examples/find_examples.py --topic coarea
python3 integral-examples/find_examples.py --technique multiplicity
python3 integral-examples/find_examples.py --prerequisite dominated
python3 integral-examples/find_examples.py --text concentration
python3 integral-examples/find_examples.py --list-tags
```

Filters are case-insensitive substring matches and combine with AND. They print titles, tags, and paths relative to `integral-examples/`. The `--text` filter searches IDs, titles, paths, topics, prerequisites, and techniques; it does not search solution prose or recorded hypotheses/results. Add `--solutions` to print the recorded result. The files remain ordinary Markdown and JSON; retrieval needs no hosted service.

1.2 How to study with the collection

For each problem, identify the function, its domain, the measure, and the integrability question before computing. In a region problem, write the indicator and determine its sections. For signed integrals, examine the absolute value or the positive and negative parts. For a parametrization, compute the Jacobian and count preimages before identifying the answer with an integral over the image. For a parameter limit, write the proposed majorant explicitly.

E01–E06 cover Lebesgue integration and iterated integrals; E07–E13 cover curves, surfaces, area, coarea, and singular maps; E14–E16 cover limit and derivative exchanges and approximate identities. Cross-links in the atlas make failure cases easy to retrieve alongside their positive criteria.

1.3 Reproduce the checks and PDFs

```
python3 integral-examples/checks/check_examples.py
python3 integral-examples/checks/check_collection.py
python3 integral-examples/build.py
```

The concrete identity checker uses SymPy and mpmath. The collection checker and retrieval command use only the Python standard library. The build uses Pandoc and pdfLaTeX. See

checks/requirements.txt for the versions actually used. The checked-in PDF and LaTeX need no build tools to read.

Independent model reviewers audited every example; the ledger records the scope, findings, and revisions. Symbolic and numerical checks supplement those reviews and are **not proofs of analytic hypotheses**. No Lean verification or human peer review is claimed. General measure-theoretic theorems are cited prerequisites; this collection does not reprove all of measure theory.

All collection changes are confined to `integral-examples/` on branch `codex/integral-examples-2026-10-15`, based on the integrated research commit `e4f337c3aa52ec4c3f2aab078d7c91b32ee0015f`. Other research branches are independent. The earlier paper was not used or reconstructed. No third-party problem collection or source PDF is reproduced, and no license is assigned here to the cited sources or to unrelated repository work.

2 Problem index — without computed answers

Each entry links to a separate full solution. Read CONVENTIONS.md first for measure normalization and the exact role of almost-everywhere sections. The JSON index and retrieval command described in README.md support search by topic, prerequisite, and technique without printing answers by default.

ID	Task	Main prerequisites	Technique to look for
E01	Integrate e^{y^2} over $0 < x < y < 1$.	Measurability, Tonelli	Change order on a measurable region
E02	Construct a region whose boundary has positive area; integrate $2y$ over it.	Countable additivity, Tonelli	Explicit removed-interval sums
E03	Can every section of a Lebesgue-measurable null set be integrated? Prove the correct general statement.	Completion, Fubini	Borel representative and null envelope
E04	Compare the iterated integrals of $(x^2 - y^2)/(x^2 + y^2)^2$ on $(0, 1)^2$.	Signed integrals, Fubini	Check positive and negative parts first
E05	Sum an explicit signed matrix by rows and by columns.	Counting measure, series	Detect infinite absolute mass
E06	Evaluate and differentiate a Frullani parameter integral.	Tonelli, dominated convergence	Replace a quotient by a parameter integral
E07	Integrate the first coordinate along a parabola with respect to length.	Derivatives, area formula	Compute the speed and pull back the weight
E08	Integrate a coordinate square on a twice-traversed circle.	Curve area formula	Count multiplicity; distinguish circulation
E09	Integrate height on a paraboloid graph over the unit disk.	Gram determinant, polar coordinates	Derive the graph area factor
E10	Integrate a coordinate square over a sphere of radius R.	Surface area formula	Handle chart seams and pole exceptions
E11	Apply weighted area to a folded square parametrization.	Jacobian, area formula	Compare source integral and image multiplicity

ID	Task	Main prerequisites	Technique to look for
E12	Turn an integral on a disk into integrals over circular level sets.	Gradient, coarea	Account for a critical point before division
E13	Explain why the Cantor staircase violates a proposed derivative-area identity.	Cantor set, absolute continuity	Locate the failed Lipschitz hypothesis
E14	Track mass in concentration and escape examples when taking limits under integrals.	Pointwise convergence, DCT	Uniform integrability and spatial tightness
E15	Compare a parameter derivative with the integral of pointwise derivatives.	Difference quotients, DCT	Test local domination; distinguish derivative limits
E16	Recover a boundary value from exponential averages.	Substitution, DCT	Approximate identity and an explicit tail estimate

Suggested routes: E01–E06 for Lebesgue integration and iterated integrals; E07–E13 for geometric integration; E14–E16 for exchanges of limits, derivatives, and integrals. E03 and E04 are useful safeguards before generalizing a successful calculation. The counterexample atlas maps the safeguards to precise sufficient conditions and contains some outcome spoilers.

3 Measures, functions, and theorems used in this library

The examples are applications for an analysis student. Their proofs are written out; the general integration theorems below are prerequisites, with sources in SOURCES.md. A computation check verifies an identity, not the measurability or integrability hypotheses that justify it.

3.1 Functions and values

A function is a map with a specified domain, codomain, and rule. For example, $f : (0,1)^2 \to \mathbb{R}$ is a function, $f(x,y)$ is its value at (x,y), and $f_x : y \mapsto f(x,y)$ is a section function. With a fixed integrand and measure, the integral is a number (or an allowed extended value), not another notation for the function. A parameter integral defines a new function, for example $F : t \mapsto \int f_t \, d\mu$; $F(t)$ is its value. When a section integral is defined only almost everywhere, a measurable version is chosen before integrating that version in the outer variable.

3.2 Lebesgue and counting measures

$\mathcal{L}^n$ denotes complete n-dimensional Lebesgue measure on $\mathbb{R}^n$. A restricted integral $\int_E f \, d\mathcal{L}^n$ means integration over the measurable set E with the restricted measure. For finite-valued f defined on all of $\mathbb{R}^n$, it equals $\int f \mathbf{1}_E \, d\mathcal{L}^n$. Counting measure on $\mathbb{N} = \{1, 2, \ldots\}$ turns integration into an explicit sum. The Lebesgue and countable counting measure factors used in the product theorems here are sigma-finite. This does not assert sigma-finiteness of $\mathcal{H}^k$ on all of $\mathbb{R}^n$ when $k < n$; our curve and surface examples use finite-measure or countably covered restrictions.

For real measurable f, set $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. A finite Lebesgue integral requires

$$\int |f| \, d\mu = \int f^+ \, d\mu + \int f^- \, d\mu < \infty.$$

An extended signed integral is defined if at least one of the two terms is finite. If both are infinite, their difference is undefined. An improper limit or principal value must be named separately; its existence does not supply a signed Lebesgue integral.

3.3 Normalized Hausdorff measure

For an integer $k \geq 1$, let $\omega_k = \mathcal{L}^k(B(0,1))$. We use

$$\mathcal{H}^k(E) = \lim_{\delta \downarrow 0} \inf \left\{ \sum_{i=1}^{\infty} \omega_k \left(\frac{\operatorname{diam} U_i}{2} \right)^k : E \subseteq \bigcup_{i=1}^{\infty} U_i, \quad \operatorname{diam} U_i \leq \delta \right\}.$$

$\mathcal{H}^0$ is counting measure. This normalization makes $\mathcal{H}^k$ agree with $\mathcal{L}^k$ on every k-plane, using an isometry to identify that plane with $\mathbb{R}^k$. Thus $\mathcal{H}^1$ measures length and $\mathcal{H}^2$ measures surface area. In particular $\omega_1 = 2$ and $\omega_2 = \pi$. Some books omit the factor $\omega_k/2^k$; their numerical values must be rescaled before using these examples. All Hausdorff integrals in this library use the convention in this paragraph.

3.4 Tonelli and Fubini, including completion

For a nonnegative product-measurable function on sigma-finite measure spaces, Tonelli permits either order of integration, including the value $+\infty$. For a signed or complex product-measurable function, Fubini applies if the absolute value has finite integral. It then gives finite section integrals almost everywhere and equality of the two iterated integrals with the product integral.

The product of two complete Lebesgue sigma-algebras need not be complete. For $\mathcal{L}^{p+q}$-measurable sets or functions on $\mathbb{R}^{p+q}$, use the completed-product versions: sections need be measurable only for almost every parameter. A Borel set has Borel sections at every parameter; an arbitrary Lebesgue-measurable set need not have measurable sections at every parameter. The outer integral uses a measurable representative of the almost-everywhere section-integral function. E03 proves this distinction directly with a null envelope. No claim about all sections is inferred from completeness.

3.5 Area with multiplicity

For a Lipschitz map $F : \mathbb{R}^m \to \mathbb{R}^n$ with $m \leq n$, write

$$J_m F(x) = \sqrt{\det(DF(x)^T DF(x))}, \qquad N(F, A, z) = \#\{x \in A : F(x) = z\}.$$

The derivative and Jacobian exist almost everywhere; set the Jacobian to zero at the exceptional points where the derivative does not exist. For a measurable domain A and nonnegative Borel weight h on the source, the weighted area formula is

$$\int_A h(x) J_m F(x)\, d\mathcal{L}^m(x) = \int_{\mathbb{R}^n} \left(\sum_{x \in A \cap F^{-1}(\{z\})} h(x) \right) d\mathcal{H}^m(z).$$

The sum is an extended nonnegative sum. With $h = 1$ it counts multiplicity. With a nonnegative Borel w and $h = w \circ F$ it becomes $w(z)N(F, A, z)$, with the nonnegative-integration convention $0 \cdot \infty = 0$. An injective parametrization therefore computes the geometric scalar integral once. If

multiplicity is q almost everywhere, the parameter integral counts it q times. For signed weights require finite absolute integral with this multiplicity. A map on a bounded domain that is Lipschitz there can be extended to apply the formula; the geometric parametrizations to which we apply area are also Lipschitz on a suitable containing compact box. The Cantor staircase is an explicitly excluded counterexample.

For a graph $F(u) = (u, g(u))$ with a scalar C^1 function g, $J_m F = \sqrt{1 + |\nabla g|^2}$. This factor is derived in the graph example.

3.6 Coarea and division by a Jacobian

For a Lipschitz map $G : \mathbb{R}^n \to \mathbb{R}^m$, $n \geq m$, the coarea Jacobian is

$$J_m G(x) = \sqrt{\det(DG(x)DG(x)^T)}.$$

For nonnegative Borel h, coarea says

$$\int_{\mathbb{R}^n} h(x) J_m G(x)\, d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} \left(\int_{G^{-1}(\{t\})} h(x)\, d\mathcal{H}^{n-m}(x) \right) d\mathcal{L}^m(t).$$

Fiber statements and inner integrals have their usual almost-everywhere meaning. Here $m = 1$ gives $J_1 G = |\nabla G|$. To recover an unweighted volume integral, division by $J_m G$ is legitimate only on its positive set. Any contribution from its zero set must be handled separately. E12 identifies and disposes of its critical point explicitly. E13 shows why a singular continuous map cannot be substituted into a Lipschitz area formula.

3.7 Scalar integration and orientation

$\int_S w\, d\mathcal{H}^k$ has no orientation. A scalar curve parametrization uses speed $|\gamma'(t)|$. A vector circulation uses $\int F(\gamma(t)) \cdot \gamma'(t)\, dt$; reversing traversal changes its sign. Flux through an oriented surface is $\int_S F \cdot \nu\, d\mathcal{H}^{n-1}$ with a chosen unit normal ν. Changing ν changes flux. Neither a tangent vector nor a normal vector is part of Hausdorff measure itself.

3.8 Convergence and differentiation

The version of monotone convergence used here concerns nonnegative increasing measurable functions. Dominated convergence concerns measurable functions and requires almost-everywhere convergence and one integrable majorant for the entire sequence. Pointwise convergence, boundedness of each individual function, or a uniform bound on the integrals alone is insufficient. E14 gives concentration and escape examples with explicit sufficient repairs. E15 states and proves a local domination criterion for differentiating an integral. E16 explains how concentration can instead be used deliberately as an approximate identity.

4 E01 — Change the order to create the missing derivative

4.1 Problem

Let

$$D = \{(x, y) \in \mathbb{R}^2 : 0 < x < y < 1\}.$$

Compute $\int_D e^{y^2}\, d\mathcal{L}^2(x, y)$ without finding an antiderivative of $y \mapsto e^{y^2}$.

4.2 Data and hypotheses

The measure is two-dimensional Lebesgue measure $\mathcal{L}^2$; the one-dimensional integrals below use $\mathcal{L}^1$. Define the function on the whole plane by

$$F(x, y) = \mathbf{1}_D(x, y)e^{y^2}.$$

Here F is a function, whereas $F(x, y)$ is its value at the point (x, y). The region D is open and hence Borel measurable. The function F is Borel measurable, is nonnegative, and satisfies $0 \le F \le e\mathbf{1}_{(0,1)^2}$. Consequently $\int_{\mathbb{R}^2} |F|\, d\mathcal{L}^2 \le e < \infty$ before any interchange is made.

4.3 Solution

1. Translate the region into sections. Fix $y \in (0, 1)$. The permitted values of x are exactly $0 < x < y$, so the horizontal section has length y. Fix $x \in (0, 1)$ instead. The permitted values of y are $x < y < 1$. Thus the two descriptions are

$$D = \{(x, y) : 0 < y < 1,\ 0 < x < y\} = \{(x, y) : 0 < x < 1,\ x < y < 1\}.$$

These are descriptions of the same set, not a rule for swapping the symbols dx and dy while keeping the bounds unchanged.

2. Integrate in the useful direction. Fubini's theorem, justified by the absolute-integrability bound above, gives

$$\begin{aligned}\int_D e^{y^2}\, d\mathcal{L}^2(x, y) &= \int_0^1 \left(\int_0^y e^{y^2}\, dx \right) dy \\ &= \int_0^1 y e^{y^2}\, dy.\end{aligned}$$

In the inner integral, y is fixed, so e^{y^2} is constant as a function of x. Multiplying that value by the section length produces the factor y.

3. Finish with a one-variable substitution. Put $u = y^2$, so $du = 2y\, dy$. Then

$$\int_0^1 y e^{y^2}\, dy = \frac{1}{2} \int_0^1 e^u\, du = \boxed{\frac{e - 1}{2}}.$$

Equivalently, this also evaluates the nested expression

$$\int_0^1 \left(\int_x^1 e^{y^2}\, dy \right) dx = \frac{e - 1}{2}.$$

The inner integral in this last expression is perfectly well-defined; lacking an elementary antiderivative does not make a function nonintegrable.

4. A sum check. Uniform convergence of the exponential power series on $[0,1]$ permits termwise integration here and gives

$$\int_0^1 ye^{y^2}\,dy = \sum_{k=0}^{\infty} \frac{1}{k!}\int_0^1 y^{2k+1}\,dy = \sum_{k=0}^{\infty} \frac{1}{2(k+1)k!} = \frac{1}{2}\sum_{j=1}^{\infty}\frac{1}{j!}.$$

The last series is $(e-1)/2$. Alternatively, the nonnegative partial sums and monotone convergence justify this same termwise calculation.

4.4 Why the theorem applies

Both factor measures are σ-finite. The region indicator makes the varying domain into a measurable function on a fixed product space. The bound $|F| \le e\mathbf{1}_{(0,1)^2}$ verifies Fubini's absolute-integrability hypothesis independently of the answer. Because F is nonnegative, Tonelli also allows either order before finiteness is known. Every section in this example is Borel measurable; the weaker almost-everywhere assertion for arbitrary Lebesgue-measurable functions is discussed in E03.

4.5 Pitfalls

- Keeping $x < y < 1$ as the inner x-bounds is meaningless: the bounds must describe the variable being integrated.
- The integral $\int_D e^{y^2}\,d\mathcal{L}^2$ integrates a function over a set. It does not integrate the set or identify a function with one of its values.
- Replacing strict inequalities by non-strict ones is harmless here because the added line segments have $\mathcal{L}^2$-measure zero. E02 explains why such a boundary argument needs verification for a general region.

4.6 Reusable technique

Tags: `region-sections`, `Fubini`, `Tonelli`, `indicator-extension`, `order-selection`, `power-series`.

When an inner primitive looks unhelpful, write both section descriptions explicitly. Integrating in the other direction can supply the derivative factor needed for a substitution. First verify measurability and nonnegativity or absolute integrability.

4.7 Sources

- John K. Hunter, *Measure Theory*, Theorem 5.18 (Fubini), printed p. 61; Theorem 4.6 (monotone convergence). Source ID: `hunter-measure`. The region, computation, and sum check above are worked out directly.

5 E02 — A measurable region whose entire boundary has positive area

5.1 Problem

Construct a closed set $C \subset [0,1]$ with empty interior and $\mathcal{L}^1(C) = 1/2$. For $E = C \times [0,1]$, determine its boundary and compute

$$\int_E 2y \, d\mathcal{L}^2(x,y).$$

Explain why simply discarding the boundary gives the wrong answer and why the zero extension of this integrand is not Riemann integrable on $[0,1]^2$.

5.2 Data and hypotheses

All measures are Lebesgue measures $\mathcal{L}^1$ and $\mathcal{L}^2$. Start with $C_0 = [0,1]$. At stage $n \geq 1$, remove from the middle of each of the 2^{n-1} closed intervals in C_{n-1} one open interval of length 4^{-n}. Let C_n be the remaining union of 2^n closed intervals and put

$$C = \bigcap_{n=0}^{\infty} C_n.$$

This is the Smith–Volterra construction specified by these exact lengths. The sets and integrands are explicit except for the infinite intersection defining C; no nonmeasurable choice is involved.

5.3 Solution

1. Check that every removal fits. All component intervals of C_n have equal length

$$\ell_n = 2^{-n-1} + 2^{-2n-1}.$$

For $n = 0$ this is 1. If the formula holds at stage $n-1$, the next component length is

$$\frac{\ell_{n-1} - 4^{-n}}{2} = 2^{-n-1} + 2^{-2n-1} > 0.$$

In particular the removed middle interval is strictly shorter than its parent, so the construction is possible at every stage.

2. Compute the measure by an explicit sum. Removed intervals from different stages are disjoint. Their total length through stage n is

$$\sum_{k=1}^{n} 2^{k-1} 4^{-k} = \frac{1}{2} \sum_{k=1}^{n} 2^{-k} = \frac{1}{2}(1 - 2^{-n}).$$

Therefore

$$\mathcal{L}^1(C_n) = \frac{1}{2} + 2^{-n-1}, \qquad \mathcal{L}^1(C) = \lim_{n\to\infty} \mathcal{L}^1(C_n) = \frac{1}{2}.$$

The limit follows from continuity from above of a measure, since C_n decreases and $\mathcal{L}^1(C_0) = 1 < \infty$.

3. Prove the interior is empty. Suppose C contained a nondegenerate interval J. For every n, the connected set J would lie in one component interval of C_n, because distinct components are separated by a removed open interval. Hence the length of J would be at most ℓ_n for every n. But $\ell_n \to 0$, a contradiction. Since C is closed and has empty interior, its complement is dense in $\mathbb{R}$.

4. Find the boundary of E. The set $E = C \times [0,1]$ is closed. It has empty interior: every open ball about a point of E contains a point whose first coordinate lies outside C. Thus

$$\partial E = \overline{E} \setminus E^\circ = E, \qquad \mathcal{L}^2(\partial E) = \mathcal{L}^2(E) = \frac{1}{2}.$$

The last equality uses the measurable-rectangle formula $\mathcal{L}^2(C \times [0,1]) = \mathcal{L}^1(C)\mathcal{L}^1([0,1])$.

5. Integrate. Extend the integrand to the square as $F(x,y) = 2y\mathbf{1}_C(x)$. It is Borel measurable and $0 \le F \le 2$, so Fubini applies:

$$\begin{aligned}\int_E 2y\, d\mathcal{L}^2(x,y) &= \int_0^1 \mathbf{1}_C(x) \left(\int_0^1 2y\, dy \right) dx \\ &= \int_0^1 \mathbf{1}_C(x)\, dx = \boxed{\frac{1}{2}}.\end{aligned}$$

Deleting the boundary would leave the empty set $E \setminus \partial E$, whose integral is 0, not $1/2$.

6. Compare with Riemann integration directly. Take any rectangular partition of $[0,1]^2$. Every subrectangle of positive width contains a point whose first coordinate is outside C. Since $F \ge 0$, its infimum on each such rectangle is 0. Every lower Darboux sum is therefore 0. Every upper Darboux step function is at least F, so its integral, equal to the corresponding upper sum, is at least $\int F\, d\mathcal{L}^2 = 1/2$. Upper and lower sums cannot approach a common value. Thus F is not Riemann integrable on the square, although it is bounded and Lebesgue integrable.

5.4 Why the theorem applies

Closedness establishes Borel measurability. Continuity from above needs the finite measure of the initial set; that hypothesis was checked. For the integral, the bounded Borel function F on a finite-measure square is absolutely integrable, so Fubini applies. No assumption that the boundary is null is required by Lebesgue integration.

5.5 Pitfalls

- "Boundary" does not imply "measure zero." Here the entire positive-area set is its boundary.
- Empty interior does not imply zero measure.
- A bounded measurable function need not be Riemann integrable. The direct Darboux argument identifies the actual obstruction in this example.
- The stage-n removal length 4^{-n} is the length of **each** of 2^{n-1} gaps, not their combined length.

5.6 Reusable technique

Tags: `fat-Cantor-set`, `positive-boundary-measure`, `explicit-series`, `continuity-from-above`, `Fubini`, `Riemann-versus-Lebesgue`.

For a complicated closed region, calculate removed measure with a disjoint sum, check interior separately, and integrate through an indicator. Treat a proposed boundary deletion as a measure claim that needs proof.

5.7 Sources

- John K. Hunter, *Measure Theory*, Proposition 1.12 (continuity of measures), Theorem 5.14 (product measure), and Theorem 5.18 (Fubini). Source ID: `hunter-measure`. The construction, interval-length formula, boundary proof, and Darboux argument are supplied here.

6 E03 — A null measurable set with one nonmeasurable section

6.1 Problem

Find a Lebesgue-measurable set $E \subset \mathbb{R}^2$ for which the vertical section

$$E_x = \{y \in \mathbb{R} : (x, y) \in E\}$$

is not Lebesgue measurable for one value of x. Reconcile this with Fubini's theorem, and prove the correct almost-everywhere statement for sections of arbitrary Lebesgue-measurable sets.

6.2 Data and hypotheses

Use complete Lebesgue measures $\mathcal{L}^1$ and $\mathcal{L}^2$. Choose a non-Lebesgue-measurable set $V \subset [0, 1]$ and define

$$E = \{0\} \times V.$$

Such a V exists under the usual axiom of choice. For completeness, choose one representative from each equivalence class in $[0, 1]$ under $s \sim t$ when $s - t \in \mathbb{Q}$. The translates $V + q$, for $q \in \mathbb{Q} \cap [-1, 1]$, are disjoint, cover $[0, 1]$, and lie in $[-1, 2]$. If V were measurable with measure zero, their countable union could not cover $[0, 1]$. If it had positive measure, countably many disjoint translates would have infinite measure inside $[-1, 2]$. These two alternatives prove nonmeasurability.

6.3 Solution

1. Joint measurability does hold. The set E lies in the Borel set $N = \{0\} \times [0, 1]$. For every $\varepsilon > 0$,

$$N \subset (-\varepsilon, \varepsilon) \times [0, 1], \qquad \mathcal{L}^2(N) \le 2\varepsilon.$$

Hence $\mathcal{L}^2(N) = 0$. Completeness of $\mathcal{L}^2$ implies that its subset E is Lebesgue measurable and has measure zero. Thus the function $\mathbf{1}_E$ is Lebesgue measurable and belongs to $L^1(\mathbb{R}^2)$, with integral 0.

2. Inspect each section. At $x = 0$, the section is $E_0 = V$, which is not Lebesgue measurable. For every $x \neq 0$, $E_x = \varnothing$, so its measure and the integral of its indicator are 0. In particular

$$y \longmapsto \mathbf{1}_E(0, y) = \mathbf{1}_V(y)$$

is a nonmeasurable section function. One must not write its Lebesgue integral as though it were defined.

For comparison, every horizontal section is either $\{0\}$ or $\varnothing$, so it is measurable and has measure zero. In this direction the section-measure function is identically 0, even though deciding which of the two section sets occurs depends on membership in V.

3. Interpret the iterated integral correctly. On the full-measure set $\mathbb{R} \setminus \{0\}$ the vertical inner integral exists and equals 0. Extend that outer integrand by the value 0 at the exceptional point. Its integral is 0, agreeing with $\int \mathbf{1}_E \, d\mathcal{L}^2 = 0$. This is an extension of an almost-everywhere defined inner integral; it is not an assertion that the original inner integral existed at $x = 0$.

4. Prove the general section statement. Let $A \subset \mathbb{R}^{m+n}$ be Lebesgue measurable. Borel regularity and completion provide Borel sets $B, N \subset \mathbb{R}^{m+n}$ such that

$$A \triangle B \subset N, \qquad \mathcal{L}^{m+n}(N) = 0.$$

Here $\triangle$ denotes symmetric difference. For every $x \in \mathbb{R}^m$, the sets B_x and N_x are Borel subsets of $\mathbb{R}^n$: the section is the preimage under the continuous map $y \mapsto (x, y)$. Since $\mathbf{1}_N \in L^1$, Fubini applied to this Borel function gives

$$\int_{\mathbb{R}^m} \mathcal{L}^n(N_x) \, d\mathcal{L}^m(x) = 0.$$

Its nonnegative integrand is therefore zero outside a measurable $\mathcal{L}^m$-null set Z. For $x \notin Z$,

$$A_x \triangle B_x \subset N_x, \qquad \mathcal{L}^n(N_x) = 0.$$

Completeness of $\mathcal{L}^n$ now implies that A_x is Lebesgue measurable and has the same measure as B_x. Thus **vertical sections are Lebesgue measurable for almost every** x. Interchanging the coordinate blocks proves the corresponding horizontal assertion. This argument does not prove measurability at parameters in Z, and the example shows why it cannot.

6.4 Why the theorem applies

In Step 4, Fubini is applied to the Borel null envelope N, whose indicator is absolutely integrable. It is not prematurely applied to a possibly nonmeasurable section of A. For an L^1 Lebesgue-measurable function, completed-measure Fubini similarly gives measurable, integrable sections almost everywhere and equality after the almost-everywhere defined outer integrand is extended on a null set.

A set in the uncompleted product σ-algebra $\mathcal{L}(\mathbb{R}^m) \otimes \mathcal{L}(\mathbb{R}^n)$ has measurable sections at **every** parameter. Indeed, sets with measurable sections form a σ-algebra containing all measurable rectangles. The example E therefore belongs to the completed planar Lebesgue σ-algebra but not to that uncompleted product σ-algebra.

6.5 Pitfalls

- "Jointly Lebesgue measurable" does not imply that every section is Lebesgue measurable.
- Completeness of each factor measure does not make the uncompleted product measure complete.
- Changing a jointly measurable function on a planar null set can damage exceptional one-dimensional sections. It does not change the joint Lebesgue integral.
- This construction requires a nonmeasurable set; it is not an explicit numerical algorithm for testing membership in V.

6.6 Reusable technique

Tags: `exceptional-sections`, `completion`, `Borel-representative`, `null-envelope`, `Fubini`, `Vitali-set`.

To justify a section argument for completed measures, replace the joint object by a Borel representative, control the difference inside a Borel null envelope, and use Fubini on that envelope. Record the exceptional parameter set rather than suppressing it.

6.7 Sources

- John K. Hunter, *Measure Theory*, Example 5.20 and Theorem 5.21, printed p. 62; Theorems 2.27–2.28 (Borel regularity and completion). Source ID: `hunter-measure`. The null-set construction is the classical completion example; the representative-and-envelope proof and its integral interpretation are detailed above.

7 E04 — Two finite iterated integrals, but no signed Lebesgue integral

7.1 Problem

On $Q = (0,1)^2$, define the real-valued function

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}.$$

Compute both iterated integrals. Determine whether $\int_Q f\, d\mathcal{L}^2$ exists as a signed Lebesgue integral, and explain exactly which hypothesis prevents use of signed Fubini.

7.2 Data and hypotheses

The measure on Q is the restriction of planar Lebesgue measure $\mathcal{L}^2$. Inner and outer one-variable integrals use $\mathcal{L}^1$. The function f is continuous on the open square, so it is Borel measurable. Its singularity is approached as $(x,y) \to (0,0)$, although that point is outside Q.

For a real-valued measurable function, put $f^+ = \max(f,0)$ and $f^- = \max(-f,0)$. The signed Lebesgue integral is $\int f^+ - \int f^-$ only when these two quantities are not both infinite. A finite signed Lebesgue integral requires $\int |f| < \infty$.

7.3 Solution

1. Integrate with respect to y first. Fix $x \in (0,1)$. As a function of $y \in [0,1]$, the formula extends continuously and is bounded, so this section is absolutely integrable. Direct differentiation gives

$$\frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2}\right) = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

Consequently

$$\int_0^1 f(x,y)\,dy = \left[\frac{y}{x^2+y^2}\right]_{y=0}^{y=1} = \frac{1}{1+x^2},$$

and the resulting outer function is integrable:

$$\int_0^1 \left(\int_0^1 f(x,y)\,dy\right) dx = \int_0^1 \frac{dx}{1+x^2} = \boxed{\frac{\pi}{4}}.$$

2. Integrate with respect to x first. For each fixed $y \in (0,1)$, the section is again continuous and bounded on $[0,1]$. This time

$$\frac{\partial}{\partial x}\left(-\frac{x}{x^2+y^2}\right) = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

Thus

$$\int_0^1 f(x,y)\,dx = -\frac{1}{1+y^2},$$

and

$$\int_0^1 \left(\int_0^1 f(x,y)\,dx\right) dy = -\int_0^1 \frac{dy}{1+y^2} = \boxed{-\frac{\pi}{4}}.$$

Every section used here is integrable, not just almost every section. That still does not establish joint absolute integrability.

3. Prove both sign parts have infinite integral by explicit disjoint rectangles. If $0 < y < x/2$, then

$$x^2 - y^2 \geq \frac{3}{4}x^2, \qquad (x^2+y^2)^2 \leq \frac{25}{16}x^4, \qquad f(x,y) \geq \frac{12}{25x^2} > 0.$$

For each integer $n \geq 0$, take

$$R_n = \left(2^{-n-1}, 2^{-n}\right) \times \left(0, 2^{-n-2}\right).$$

These rectangles are pairwise disjoint and lie in $\{0 < y < x/2\} \cap Q$. On R_n, $x < 2^{-n}$, so

$$f^+(x,y) \geq \frac{12}{25} 2^{2n}, \qquad \mathcal{L}^2(R_n) = 2^{-2n-3}.$$

For every positive integer N, monotonicity of the nonnegative integral yields

$$\begin{aligned} \int_Q f^+ \, d\mathcal{L}^2 &\geq \sum_{n=0}^{N-1} \frac{12}{25} 2^{2n} \mathcal{L}^2(R_n) \\ &= \sum_{n=0}^{N-1} \frac{3}{50} = \frac{3N}{50}. \end{aligned}$$

Letting $N \to \infty$ proves $\int_Q f^+ = \infty$. Since $f(y,x) = -f(x,y)$, the transposed rectangles give the identical bound for f^-. Hence

$$\int_Q f^+ \, d\mathcal{L}^2 = \int_Q f^- \, d\mathcal{L}^2 = \infty, \qquad \int_Q |f| \, d\mathcal{L}^2 = \infty.$$

The expression $\int_Q f$ is therefore **undefined as a signed Lebesgue integral**, rather than equal to either iterated value or to zero.

4. Identify another prescription, without confusing it with a Lebesgue integral. On $(\varepsilon, 1)^2 \subset Q$, where $0 < \varepsilon < 1$, the function is bounded and antisymmetric under exchanging coordinates. Fubini is valid on that truncated square, and

$$\int_{(\varepsilon,1)^2} f \, d\mathcal{L}^2 = 0.$$

Thus this particular symmetric limiting prescription gives zero. The disagreement with the two iterated values records dependence on the truncation procedure; it supplies no value for the undefined signed Lebesgue integral.

7.4 Why the theorem applies

The one-dimensional computations use the fundamental theorem of calculus on continuous sections. Fubini applies on each truncated square because the integrand is bounded there. It does **not** apply on Q: joint absolute integrability fails by Step 3. Tonelli remains valid separately for the nonnegative functions f^+, f^-, and $|f|$, with value ∞ where indicated. It never legitimizes subtracting $\infty - \infty$.

7.5 Pitfalls

- Finite iterated **signed** integrals do not replace the hypothesis $\int_Q |f| < \infty$.
- Cancellation inside each section can conceal infinite joint positive and negative masses.
- Symmetry gives an integral of zero only after the existence of that integral has been justified, or for a specifically stated finite truncation.
- Defining a value for f at the missing origin would not repair the divergence in its neighborhood.

7.6 Reusable technique

Tags: `signed-Fubini-failure`, `absolute-integrability`, `positive-negative-parts`, `dyadic-rectangles`, `cancellation`, `truncation-dependence`.

Before exchanging signed integrals, estimate $|f|$ or its sign parts. A sequence of disjoint regions each carrying a fixed positive mass proves divergence with an explicit sum, even when direct polar integration is inconvenient. A sufficient repair is to impose $f \in L^1(Q)$; local boundedness away from the singularity is insufficient.

7.7 Sources

- John K. Hunter, *Measure Theory*, Definition 4.8 (signed integral), Theorem 5.18 (Fubini), and Theorem 5.21 (completed Lebesgue measures). Source ID: `hunter-measure`. The primitive identities and dyadic-rectangle lower bounds are checked directly above.

8 E05 — Two finite iterated sums and no signed double integral

8.1 Problem

Let $\mathbb{N} = \{1, 2, \ldots\}$ and define the function $a : \mathbb{N}^2 \to \mathbb{R}$ by

$$a(n,m) = \begin{cases} 1 & m = n, \\ -1 & m = n+1, \\ 0 & \text{otherwise.} \end{cases}$$

Compute both iterated sums. Explain exactly why their disagreement does not contradict Fubini's theorem, and what square partial sums do.

8.2 Data and hypotheses

Each copy of $\mathbb{N}$ has the power-set σ-algebra and counting measure $\#$; the product is counting measure on $\mathbb{N}^2$. These spaces are σ-finite because $\mathbb{N}$ is a countable union of finite sets. Every function here is measurable. Write $a_{nm} = a(n,m)$ for a **value** of the function a. A signed Lebesgue integral is defined only if at least one of $\int a^+ \, d(\# \otimes \#)$ and $\int a^- \, d(\# \otimes \#)$ is finite; an integrable real-valued function must satisfy $\int |a| < \infty$.

8.3 Solution

1. Sum each row first. For every fixed n, only two entries are nonzero:

$$\sum_{m=1}^{\infty} a_{nm} = a_{nn} + a_{n,n+1} = 1 - 1 = 0.$$

Consequently

$$\sum_{n=1}^{\infty} \left(\sum_{m=1}^{\infty} a_{nm} \right) = \sum_{n=1}^{\infty} 0 = 0.$$

Every inner sum is absolutely convergent, and the resulting outer series is absolutely convergent.

2. Sum each column first. Column $m = 1$ contains only $a_{11} = 1$. For every $m \geq 2$, its nonzero entries are $a_{mm} = 1$ and $a_{m-1,m} = -1$. Thus

$$\sum_{n=1}^{\infty} a_{n1} = 1, \qquad \sum_{n=1}^{\infty} a_{nm} = 0 \quad (m \geq 2),$$

and

$$\sum_{m=1}^{\infty} \left(\sum_{n=1}^{\infty} a_{nm} \right) = 1 + \sum_{m=2}^{\infty} 0 = 1.$$

Again, every inner sum and the resulting outer series converge absolutely. What fails is absolute summability **before** cancellation:

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} |a_{nm}| = \sum_{n=1}^{\infty} 2 = +\infty.$$

Indeed the positive diagonal and negative superdiagonal each contain infinitely many entries of magnitude 1, so $\int a^+ = \int a^- = +\infty$. The signed double integral $\int a \, d(\# \otimes \#)$ is undefined; neither iterated answer can be identified with it.

3. Inspect finite rectangles. For positive integers N, M,

$$S_{N,M} = \sum_{n=1}^{N} \sum_{m=1}^{M} a_{nm} = \min(N, M) - \min(N, M-1).$$

Therefore $S_{N,N} = 1$, whereas $S_{N,N+1} = 0$. There is no limit as $N, M \to \infty$ independently. Taking $M \to \infty$ first gives 0; taking $N \to \infty$ first gives 1. A square exhaustion happens to select the second value, but it does not define a signed Lebesgue integral.

8.4 Why the theorem applies

Tonelli's theorem applies to each of $|a|$, a^+ and a^-, because they are nonnegative measurable functions on σ-finite product spaces. It correctly gives infinite integrals. Signed Fubini requires $\sum_{n,m} |a_{nm}| < \infty$, which has explicitly failed. If instead a doubly indexed real or complex family b_{nm} satisfies that bound, Fubini permits interchanging its sums and identifies both with the product integral.

8.5 Pitfalls

- Absolute convergence of every row, every column, and even both series of already-summed values does not establish $\sum_{n,m} |a_{nm}| < \infty$.
- This is an iterated-summation counterexample, not a conditionally Lebesgue-integrable function. For real or complex functions, Lebesgue integrability entails absolute integrability. Ordinary conditionally convergent series use an ordering, which counting-measure integration does not supply.
- Never replace the undefined expression $+\infty - (+\infty)$ by zero after pairing entries.

8.6 Reusable technique

Before exchanging signed sums or integrals, apply Tonelli to the absolute value. If it diverges, inspect positive and negative parts and the actual limiting convention. Tags: `counting-measure`, `signed-fubini`, `absolute-integrability`, `order-of-summation`, `counterexample`.

8.7 Sources

- John K. Hunter, *Measure Theory*, Theorem 5.18 and Example 5.19 (Fubini and absolutely convergent double series), and Chapter 4 (signed integration). Source ID: `hunter-measure`.
- The matrix, explicit rectangular calculation, and exposition above were worked out for this collection; the source supports the theorem, not a copied exercise.

9 E06 — A logarithm from a positive double integral

9.1 Problem

For $a, b > 0$, calculate

$$F(a,b) = \int_0^\infty \frac{e^{-ax} - e^{-bx}}{x}\,dx$$

as a Lebesgue integral. Justify both the exchange of integrations and differentiation with respect to b.

9.2 Data and hypotheses

The integration variable x carries one-dimensional Lebesgue measure λ^1 on $(0,\infty)$. A parameter variable t carries λ^1 on a finite interval. The expression $(e^{-ax} - e^{-bx})/x$ is the value at x of a continuous function on $(0,\infty)$. Positivity of both parameters is essential for the tail estimate below. No value at $x = 0$ is needed; assigning the limiting value $b - a$ there would not change the integral.

9.3 Solution

1. Represent the cancellation before integrating. Suppose first $a \le b$. The fundamental theorem of calculus in the parameter gives, for $x > 0$,

$$\frac{e^{-ax} - e^{-bx}}{x} = \int_a^b e^{-tx}\,dt \ge 0.$$

The function $(x,t) \mapsto e^{-tx}$ is continuous and nonnegative. Tonelli therefore allows an exchange before finiteness has been established:

$$\begin{aligned} F(a,b) &= \int_0^\infty \int_a^b e^{-tx}\,dt\,dx \\ &= \int_a^b \int_0^\infty e^{-tx}\,dx\,dt = \int_a^b \frac{1}{t}\,dt = \log b - \log a. \end{aligned}$$

The inner integral is $1/t$ because $t \ge a > 0$. The resulting logarithm is finite, so this also proves integrability. If $a > b$, apply this computation to $F(b,a)$ and use the pointwise negative sign; if $a = b$, the integrand is identically zero. In all cases,

$$\boxed{F(a,b) = \log(b/a).}$$

2. Check absolute integrability directly. Let $c = \min(a,b) > 0$ and $d = |b - a|$. The same parameter representation with the endpoints ordered gives

$$\left|\frac{e^{-ax} - e^{-bx}}{x}\right| \le de^{-cx} \le d \qquad (x > 0).$$

Thus the integral of the absolute value on $(0,1)$ is at most d. On $(1,\infty)$ one may also use

$$\left|\frac{e^{-ax}-e^{-bx}}{x}\right| \le e^{-ax}+e^{-bx},$$

whose integral is finite. This makes the endpoint hypotheses visible without relying on the final formula.

3. Differentiate under the integral with a local bound. Fix $a>0$ and $b_0>0$, and write $q_b(x)=(e^{-ax}-e^{-bx})/x$. Its parameter derivative is

$$\partial_b q_b(x) = e^{-bx}.$$

For $b \in [b_0/2, 3b_0/2]$,

$$|\partial_b q_b(x)| \le e^{-b_0 x/2}, \qquad \int_0^\infty e^{-b_0 x/2}\,dx = 2/b_0 < \infty.$$

For sufficiently small nonzero h, the ordinary mean value theorem bounds the difference quotient $(q_{b_0+h}(x)-q_{b_0}(x))/h$ by this same integrable function. These quotients converge pointwise to $e^{-b_0 x}$. Dominated convergence yields

$$\partial_b F(a,b_0) = \int_0^\infty e^{-b_0 x}\,dx = 1/b_0,$$

in agreement with the logarithm. This is a proof of the exchange, rather than a deduction from the already-known answer.

9.4 Why the theorem applies

Tonelli applies on the σ-finite Lebesgue product $(0,\infty)\times[a,b]$ to a nonnegative Borel function. Once its integral is finite, signed Fubini would also be available. For differentiation, the difference quotients are measurable, converge pointwise, and have a common L^1 majorant on a neighborhood of the chosen positive parameter; Hunter's dominated convergence theorem applies. The same argument works near any $b_0>0$; a uniform bound over all $b>0$ is unnecessary.

9.5 Pitfalls

- The two terms e^{-ax}/x and e^{-bx}/x each have infinite integrals near zero. Writing their separate integrals and subtracting creates an undefined $\infty-\infty$.
- If $a=0<b$, then $(1-e^{-bx})/x$ has a divergent tail, so the stated finite formula does not extend to $a=0$.
- A bound on q_b alone does not justify differentiating it. The proof controls its difference quotients through $\partial_b q_b$.

9.6 Reusable technique

Turn a difference quotient or cancellation into an integral over a positive parameter kernel. Apply Tonelli before evaluating, then use a local integrable derivative bound for parameter differentiation. Tags: `frullani`, `tonelli`, `parameter-integral`, `differentiate-under-integral`, `cancellation`.

9.7 Sources

- John K. Hunter, *Measure Theory*, Theorem 4.24 (dominated convergence) and Chapter 5 (product integration). Source ID: `hunter-measure`.
- Keith Conrad, *Differentiating under the integral sign*, Theorem 12.3, for a classical sufficient differentiation theorem. The measurable version used here follows from the displayed difference-quotient proof. Source ID: `conrad-diff`.

10 E07 — A weighted integral on a parabolic arc

10.1 Problem

Let $C = \{(t, t^2) : 0 \le t \le 1\}$. Compute the integral of the coordinate function $f : C \to \mathbb{R}$, defined by $f(x, y) = x$, with respect to arc length. Also compute the length of C.

10.2 Data and hypotheses

The parametrization is $\gamma : [0, 1] \to \mathbb{R}^2$, $\gamma(t) = (t, t^2)$. The curve C is compact, and f is continuous and nonnegative on C.

Throughout, $\mathcal{L}^k$ denotes Lebesgue measure. For an integer $k \ge 1$, use the normalized Hausdorff measure

$$\mathcal{H}^k(A) = \lim_{\delta \downarrow 0} \inf \left\{ \sum_{j=1}^{\infty} \frac{\alpha_k}{2^k} (\operatorname{diam} U_j)^k : A \subseteq \bigcup_{j=1}^{\infty} U_j,\ \operatorname{diam} U_j < \delta \right\},$$

where $\alpha_k = \mathcal{L}^k(B^k(0, 1))$. Thus $\mathcal{H}^k$ on a k-plane agrees with k-dimensional Lebesgue measure, and $\mathcal{H}^0$ is counting measure. In particular, $\alpha_1 = 2$, so $\mathcal{H}^1$ here is ordinary arc length. The measure being integrated is the restriction $\mathcal{H}^1 \mid C$, not two-dimensional area measure.

10.3 Solution

Differentiate the parametrization:

$$\gamma'(t) = (1, 2t), \qquad J_1\gamma(t) = |\gamma'(t)| = \sqrt{1 + 4t^2}.$$

The first coordinate proves injectivity: if $\gamma(s) = \gamma(t)$, then $s = t$. The weighted area formula therefore gives

$$\int_C f \, d\mathcal{H}^1 = \int_0^1 f(\gamma(t)) |\gamma'(t)| \, dt = \int_0^1 t\sqrt{1 + 4t^2} \, dt.$$

Here f is the function; $f(\gamma(t)) = t$ is its value at the point $\gamma(t)$. With $u = 1 + 4t^2$ and $du = 8t \, dt$,

$$\int_0^1 t\sqrt{1 + 4t^2} \, dt = \frac{1}{8} \int_1^5 u^{1/2} \, du = \frac{1}{12} [u^{3/2}]_1^5 = \boxed{\frac{5\sqrt{5} - 1}{12}}.$$

For length, replace f by the constant function 1:

$$\mathcal{H}^1(C) = \int_0^1 \sqrt{1 + 4t^2}\, dt.$$

Put $s = 2t$. The identity

$$\frac{d}{ds}\left(\frac{1}{2}\left(s\sqrt{1+s^2} + \operatorname{arsinh} s\right)\right) = \sqrt{1+s^2}$$

yields

$$\mathcal{H}^1(C) = \frac{\sqrt{5}}{2} + \frac{1}{4}\operatorname{arsinh} 2 = \boxed{\frac{\sqrt{5}}{2} + \frac{1}{4}\log(2+\sqrt{5})}.$$

10.4 Why the theorem applies

The map γ is C^1 on a neighborhood of $[0,1]$ and has bounded derivative there, hence is Lipschitz on this interval. Its domain is Borel. Its multiplicity is exactly one on C. The weighted area formula applies to the nonnegative Borel weight $f \circ \gamma$; the explicit integral is finite since $0 \le t\sqrt{1+4t^2} \le \sqrt{5}$. Taking the constant weight also proves that C has finite $\mathcal{H}^1$ measure.

10.5 Pitfalls

- Integrating $t\,dt$ gives $1/2$, but this uses horizontal distance instead of arc length. The speed factor is necessary.
- The expression $\int_C f\, d\mathcal{L}^2$ would be zero: C has two-dimensional Lebesgue measure zero. A Lipschitz arc can be covered by N disks of radius proportional to $1/N$, with total area tending to zero.
- Reversing the traversal leaves this scalar Hausdorff integral unchanged. A vector circulation would include a signed tangent vector and can change sign.

10.6 Reusable technique

For an injective Lipschitz curve, compose the scalar weight with the parametrization, then multiply by speed. Coordinate weights often turn a square-root integral into a direct substitution.

Tags: `hausdorff-normalization`, `weighted-arc-length`, `injective-area-formula`, `substitution`, `coordinate-moment`.

10.7 Sources

- Leon Simon, *Introduction to Geometric Measure Theory* (2018), Chapter 1, §2, definitions 2.1–2.2 and Theorem 2.6 (normalization); Chapter 2, §3, Theorem 3.3, Remark 3.4(2), and Example 3.5(1) (weighted area formula and arc length). The specific curve and computations above are worked out here. Source ID: `simon-gmt`.

11 E08 — Traversing a circle twice: multiplicity and orientation

11.1 Problem

Let $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ and let $w : C \to \mathbb{R}$ be the function $w(x, y) = x^2$. Compare

$$\int_C w \, d\mathcal{H}^1 \quad \text{and} \quad \int_0^{4\pi} w(\gamma(t))|\gamma'(t)| \, dt, \qquad \gamma(t) = (\cos t, \sin t).$$

Then compare these scalar integrals with the circulation of the vector field $F(x, y) = (-y, x)$.

11.2 Data and hypotheses

Use normalized Hausdorff measure: $\mathcal{H}^k$ has covering cost $\alpha_k(\operatorname{diam} U/2)^k$, where α_k is the volume of the unit ball in $\mathbb{R}^k$. It agrees with $\mathcal{L}^k$ on k-planes; $\mathcal{H}^0$ counts points. Thus $\mathcal{H}^1 \mid C$ is ordinary length measure. The functions w and F are continuous, and $0 \le w \le 1$ on C.

The map γ is Lipschitz on $A = [0, 4\pi]$. Its image is the geometric set C, which has no record of how many times it was traversed.

11.3 Solution

First compute the parameter integral. Since

$$\gamma'(t) = (-\sin t, \cos t), \qquad |\gamma'(t)| = 1, \qquad w(\gamma(t)) = \cos^2 t,$$

we obtain

$$\int_0^{4\pi} w(\gamma(t))|\gamma'(t)| \, dt = \int_0^{4\pi} \frac{1 + \cos(2t)}{2} \, dt = \boxed{2\pi}.$$

This is a traversal integral. To understand its relation to the geometric integral, define

$$N(q) = \mathcal{H}^0(\{t \in A : \gamma(t) = q\}).$$

If $q \in C \setminus \{(1, 0)\}$, there are exactly two preimages, one in $(0, 2\pi)$ and one in $(2\pi, 4\pi)$. At $(1, 0)$ the preimages are $0, 2\pi, 4\pi$, so $N(1, 0) = 3$. Off C, $N = 0$. The single exceptional image point has $\mathcal{H}^1$ measure zero. The weighted area formula therefore reads

$$\begin{aligned} \int_0^{4\pi} w(\gamma(t))|\gamma'(t)| \, dt &= \int_C \sum_{t \in \gamma^{-1}(q) \cap A} w(q) \, d\mathcal{H}^1(q) \\ &= \int_C w(q) N(q) \, d\mathcal{H}^1(q) = 2 \int_C w \, d\mathcal{H}^1. \end{aligned}$$

Consequently,

$$\boxed{\int_C w \, d\mathcal{H}^1 = \pi}.$$

A single traversal $0 \le t \le 2\pi$ gives the same geometric answer directly: only the common endpoint is counted twice, and it is length-null.

For circulation, the integrand includes the velocity vector rather than its norm:

$$F(\gamma(t)) = (-\sin t, \cos t) = \gamma'(t), \qquad F(\gamma(t)) \cdot \gamma'(t) = 1.$$

Thus the double counterclockwise traversal has circulation

$$\int_0^{4\pi} F(\gamma(t)) \cdot \gamma'(t)\, dt = \boxed{4\pi},$$

and a single counterclockwise traversal has circulation 2π. A reversed single traversal has circulation -2π, whereas its scalar w-integral remains π. These circulation values are not integrals of the scalar function w.

Flux requires a chosen unit normal: it has the form $\int_C G \cdot n\, d\mathcal{H}^1$. For example, $G(x, y) = (x, y)$ has outward flux 2π, since the outward normal is $n(q) = q$ and $G(q) \cdot n(q) = 1$. Choosing the inward normal changes that flux to -2π. The positive measure $\mathcal{H}^1$ itself supplies neither a tangent orientation nor a normal orientation.

11.4 Why the theorem applies

The domain A is compact Borel and γ is C^1 with bounded derivative. The nonnegative weight $t \mapsto w(\gamma(t))$ is Borel. Hence the weighted area formula applies even though γ is not injective. Its fiber sum is explicit above. Integrability is also direct: the parameter integrand is bounded by 1 on a finite interval. The same formula with weight 1 shows $\mathcal{H}^1(C) = 2\pi$, which justifies all bounded scalar and flux integrals used here.

11.5 Pitfalls

- A parametrization's image alone does not determine its traversal integral. Multiplicity must be included.
- Multiplicity is a count, never a signed count in the scalar area formula. Opposite traversals add for scalar length even when their circulations cancel.
- Endpoint multiplicity is genuinely three at $(1, 0)$ for the stated closed domain. Ignoring it is justified by its null image, not by claiming it is two everywhere.

11.6 Reusable technique

Before using a noninjective parametrization, count its fibers. The area formula converts a parameter integral into an integral of an explicit sum over preimages. Treat orientation separately whenever a vector field is present.

Tags: `area-formula`, `multiplicity`, `noninjective-parametrization`, `scalar-versus-vector`, `circulation`, `flux`.

11.7 Sources

- Leon Simon, *Introduction to Geometric Measure Theory* (2018), Chapter 1, §2, definitions 2.1–2.2 and Theorem 2.6; Chapter 2, §3, Theorem 3.3 and Remark 3.4(2) (weighted fiber sum). The fiber count and vector comparisons above are direct computations. Source ID: `simon-gmt`.

12 E09 — Height integral over a paraboloid graph

12.1 Problem

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$ and

$$S = \{(x, y, x^2 + y^2) : (x, y) \in D\}.$$

For the height function $h : S \to \mathbb{R}$, $h(x, y, z) = z$, compute $\int_S h \, d\mathcal{H}^2$. Also compute the area of S.

12.2 Data and hypotheses

All measures are positive scalar measures. The normalized Hausdorff measure has covering cost $\alpha_k(\operatorname{diam} U/2)^k$, with $\alpha_k = \mathcal{L}^k(B^k(0, 1))$; it agrees with $\mathcal{L}^k$ on k-planes. Thus the cost for $\mathcal{H}^2$ is $(\pi/4)(\operatorname{diam} U)^2$, and $\mathcal{H}^0$ is counting measure.

Write $u : D \to \mathbb{R}$, $u(x, y) = x^2 + y^2$, and $\Phi : D \to \mathbb{R}^3$, $\Phi(x, y) = (x, y, u(x, y))$. The domain D is compact Borel. The map Φ is injective because its first two coordinates recover its input. The function h is continuous with $0 \le h \le 1$ on S.

12.3 Solution

Compute the derivative columns and their Gram matrix:

$$\partial_x \Phi = (1, 0, 2x), \qquad \partial_y \Phi = (0, 1, 2y),$$

$$(D\Phi)^T D\Phi = \begin{pmatrix} 1 + 4x^2 & 4xy \\ 4xy & 1 + 4y^2 \end{pmatrix}.$$

Its determinant is

$$(1 + 4x^2)(1 + 4y^2) - 16x^2y^2 = 1 + 4x^2 + 4y^2.$$

Hence the two-dimensional area factor is

$$J_2\Phi(x, y) = \sqrt{1 + 4x^2 + 4y^2} = \sqrt{1 + |\nabla u(x, y)|^2}.$$

Injectivity and the weighted area formula give

$$\int_S h \, d\mathcal{H}^2 = \int_D h(\Phi(x, y)) J_2\Phi(x, y) \, d\mathcal{L}^2(x, y) = \int_D (x^2 + y^2)\sqrt{1 + 4x^2 + 4y^2} \, dx \, dy.$$

The function h is defined on S; its pulled-back value is $h(\Phi(x,y)) = u(x,y)$. Now use polar coordinates in the planar disk, including their separate Jacobian factor r:

$$I = 2\pi \int_0^1 r^3 \sqrt{1+4r^2}\, dr.$$

For $v = 1 + 4r^2$, we have $r\, dr = dv/8$ and $r^2 = (v-1)/4$, so

$$\begin{aligned} I &= \frac{\pi}{16} \int_1^5 (v-1) v^{1/2}\, dv \\ &= \frac{\pi}{16} \left[\frac{2}{5} v^{5/2} - \frac{2}{3} v^{3/2} \right]_1^5 \\ &= \frac{\pi}{16} \left(\frac{20\sqrt{5}}{3} + \frac{4}{15} \right) = \boxed{\frac{\pi(25\sqrt{5}+1)}{60}}. \end{aligned}$$

Using weight 1 instead gives

$$\begin{aligned} \mathcal{H}^2(S) &= 2\pi \int_0^1 r\sqrt{1+4r^2}\, dr \\ &= \frac{\pi}{4} \int_1^5 v^{1/2}\, dv = \boxed{\frac{\pi(5\sqrt{5}-1)}{6}}. \end{aligned}$$

As a quick consistency check, $0 < I < \mathcal{H}^2(S)$, as expected because the height lies between 0 and 1 and is strictly between them on a set of positive area.

12.4 Why the theorem applies

On the convex disk, $|\nabla u| \le 2$, so $|u(p) - u(q)| \le 2|p-q|$ and $|\Phi(p) - \Phi(q)| \le \sqrt{5}|p-q|$. Thus Φ is Lipschitz on D. Its image is compact Borel, its multiplicity is one, and the weight is nonnegative Borel. The explicit area factor is bounded by $\sqrt{5}$, so both pulled-back integrands are integrable on D. The polar coordinate change is applied to a bounded, nonnegative measurable function; its seam and origin have planar measure zero. The area formula handles the closed disk directly, so there is no need to assume an open parameter domain or discard the boundary without explanation.

12.5 Pitfalls

- The projected integral $\int_D u\, dx\, dy = \pi/2$ differs from the surface integral. Projection omits the graph's area factor.
- There are two distinct factors after the two changes of variables: $\sqrt{1+4r^2}$ comes from graph area; r comes from planar polar coordinates.
- The cross product $(-2x, -2y, 1)$ can also compute the area factor, but only its norm enters this scalar integral. A signed vector flux would additionally require a chosen normal.

12.6 Reusable technique

For a graph $z = u(x,y)$ with one point above each base point, pull the scalar weight back and multiply by $\sqrt{1+|\nabla u|^2}$. If the domain and pulled-back integrand are radial, use polar coordinates only after forming this graph factor.

Tags: `graph-area`, `weighted-surface-integral`, `gram-determinant`, `polar-coordinates`, `injective-area-formula`, `substitution`.

12.7 Sources

- Leon Simon, *Introduction to Geometric Measure Theory* (2018), Chapter 1, §2, definitions 2.1–2.2 and Theorem 2.6; Chapter 2, §3, definition 3.2, Theorem 3.3, Remark 3.4(2), and Example 3.5(3) (graph area factor). The Gram determinant, substitutions, and numerical constants above are derived here. Source ID: `simon-gmt`.

13 E10 — A sphere's second moment and a spherical cap

13.1 Problem

For $R > 0$, let $S_R = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = R^2\}$. Compute the scalar integral of the function $h : S_R \to \mathbb{R}$, $h(x, y, z) = z^2$:

$$\int_{S_R} h \, d\mathcal{H}^2.$$

As a related application, compute the area of a cap of height $a \in [0, 2R]$ defined by $z \geq R - a$.

13.2 Data and hypotheses

Use normalized Hausdorff measure: the covering cost is $\alpha_k(\operatorname{diam} U/2)^k$, where α_k is the k-dimensional unit-ball volume. This makes $\mathcal{H}^k$ equal to $\mathcal{L}^k$ on k-planes; $\mathcal{H}^0$ is counting measure. Thus $\mathcal{H}^2$ has covering cost $(\pi/4)(\operatorname{diam} U)^2$ and is usual surface area on a smooth surface. The sphere and cap are compact Borel sets, and h is continuous and nonnegative.

13.3 Solution

Use the spherical map

$$\Phi(\theta, \varphi) = (R \sin\theta \cos\varphi, R \sin\theta \sin\varphi, R \cos\theta),$$

on $D = (0, \pi) \times (0, 2\pi)$. Its derivative columns are perpendicular, with lengths R and $R \sin\theta$. Therefore

$$J_2\Phi(\theta, \varphi) = R^2 \sin\theta.$$

The map is injective on D: z/R determines $\theta \in (0, \pi)$, and the first two coordinates then determine the unique $\varphi \in (0, 2\pi)$. Its image omits exactly the meridian

$$M = \{(R \sin\theta, 0, R \cos\theta) : 0 \leq \theta \leq \pi\},$$

including the poles. This omitted set is $\mathcal{H}^2$-null. Indeed, its displayed curve is R-Lipschitz; partitioning $[0, \pi]$ into N intervals covers M by N sets of diameter at most $R\pi/N$. Their total

normalized two-dimensional cost is at most $\pi^3 R^2/(4N)$, tending to zero, while their maximum diameter also tends to zero.

The weighted area formula now gives

$$\begin{aligned}\int_{S_R} h\, d\mathcal{H}^2 &= \int_0^{2\pi}\int_0^{\pi} R^2\cos^2\theta\, R^2\sin\theta\, d\theta\, d\varphi \\ &= 2\pi R^4 \int_0^{\pi} \cos^2\theta\sin\theta\, d\theta.\end{aligned}$$

With $v = \cos\theta$, $dv = -\sin\theta\, d\theta$,

$$\int_0^{\pi} \cos^2\theta\sin\theta\, d\theta = \int_{-1}^{1} v^2\, dv = \frac{2}{3},$$

so

$$\boxed{\int_{S_R} z^2\, d\mathcal{H}^2 = \frac{4\pi R^4}{3}}.$$

The same parametrization with weight 1 gives $\mathcal{H}^2(S_R) = 4\pi R^2$. A separate check uses coordinate symmetry: Hausdorff measure is unchanged by rotations, so the integrals of x^2, y^2, and z^2 are equal. Their sum is

$$\int_{S_R} (x^2+y^2+z^2)\, d\mathcal{H}^2 = R^2\mathcal{H}^2(S_R) = 4\pi R^4.$$

For the cap, set $\theta_0 = \arccos(1 - a/R)$. The condition $z \geq R - a$ is $0 \leq \theta \leq \theta_0$. Thus

$$\mathcal{H}^2(\{q \in S_R : z(q) \geq R - a\}) = 2\pi R^2 \int_0^{\theta_0} \sin\theta\, d\theta = \boxed{2\pi R a}.$$

This includes $a = 0$ (a single point, area zero) and $a = 2R$ (the entire sphere).

13.4 Why the theorem applies

The map Φ extends smoothly to the closed rectangle $[0,\pi]\times[0,2\pi]$, with bounded derivative, hence is Lipschitz there and on D. The restricted domain D is Borel and has multiplicity one on its image. Its Jacobian is bounded by R^2, while $h\circ\Phi$ is bounded by R^2 on a finite-measure rectangle. All scalar integrals are therefore finite. The missing meridian is null by the covering argument above. Nonnegative integrands justify the iterated integrations by Tonelli; finiteness also permits Fubini. Cap boundary latitudes are Lipschitz curves and hence $\mathcal{H}^2$-null by the same covering argument.

13.5 Pitfalls

- The factor $R^2\sin\theta$ is essential. The angular parameter rectangle carries planar measure, not surface measure.

- The closed-rectangle parametrization is not injective at the seam or poles. Restricting to D and proving that the omitted image is null resolves this explicitly.
- This scalar moment has no chosen orientation. For a vector field $G(q) = q$, the outward unit normal is $n(q) = q/R$ and the outward flux is $\int_{S_R} G \cdot n \, d\mathcal{H}^2 = R\mathcal{H}^2(S_R) = 4\pi R^3$. The inward normal reverses that flux, while the z^2 integral remains unchanged.

13.6 Reusable technique

Compute a surface Jacobian from the Gram determinant or perpendicular tangent vectors. Separate chart overlap/null-image issues from the calculation. Use rotation symmetry to check coordinate moments, and express caps through their polar angle or height.

Tags: `sphere`, `surface-moment`, `spherical-cap`, `area-formula`, `null-seam`, `rotation-symmetry`, `scalar-versus-flux`.

13.7 Sources

- Leon Simon, *Introduction to Geometric Measure Theory* (2018), Chapter 1, §2, definitions 2.1–2.2 and Theorem 2.6; Chapter 2, §3, definition 3.2, Theorem 3.3, Remark 3.4(2), and Example 3.5(2) (surface measure and area formula). The chart, null-meridian proof, moment, cap computation, and symmetry check above are derived here. Source ID: `simon-gmt`.

14 E11 — A fold counts the image twice

14.1 Problem

Let $A = [-1, 1] \times [0, 1]$ and define the map $F : A \to \mathbb{R}^2$ by

$$F(u, v) = (u^2, v).$$

Compute the image area, the integral of the area Jacobian over A, and the weighted integral

$$\int_A (u^2 + v) J_2 F(u, v) \, d\mathcal{L}^2(u, v).$$

Explain why an injective change-of-variables formula would give the wrong answer. Then recover the scalar integral of $h(x, y) = x + y$ over the image, counted once.

14.2 Data and hypotheses

The domain and target measures are completed two-dimensional Lebesgue measure $\mathcal{L}^2$. We use normalized Hausdorff measure

$$\mathcal{H}^k(E) = \lim_{\delta \downarrow 0} \inf \left\{ \sum_j \frac{\alpha_k}{2^k} (\operatorname{diam} U_j)^k : E \subseteq \bigcup_j U_j, \ \operatorname{diam} U_j < \delta \right\},$$

where α_k is the volume of the unit ball in $\mathbb{R}^k$. Thus $\mathcal{H}^k$ agrees with $\mathcal{L}^k$ on a k-plane; $\mathcal{H}^0$ is counting measure.

The set A is compact and Borel. The map F is the restriction of a smooth map, and is Lipschitz on A: $|u^2 - s^2| \le 2|u - s|$ for $u, s \in [-1, 1]$ gives $|F(u, v) - F(s, t)| \le 2|(u, v) - (s, t)|$. The scalar

function h is continuous and nonnegative on the image. All integrals computed below are finite. The multiplicity is the function

$$N(F, A, z) = \mathcal{H}^0(A \cap F^{-1}(\{z\})),$$

not the map F itself and not a chosen inverse branch.

14.3 Solution

1. Compute the image and every fibre. The image is $Q = [0,1] \times [0,1]$. For $z = (x, y) \in Q$ with $x > 0$, the fibre consists of the two distinct points

$$A \cap F^{-1}(\{(x,y)\}) = \{(\sqrt{x}, y), (-\sqrt{x}, y)\}.$$

At $x = 0$ it consists of the single point $(0, y)$, and outside Q it is empty. Consequently $N = 2$ on $(0,1] \times [0,1]$, $N = 1$ on $\{0\} \times [0,1]$, and $N = 0$ elsewhere. In particular $N = 2$ at $\mathcal{L}^2$-almost every point of Q.

2. Compute the area Jacobian. Differentiating the map gives

$$DF(u,v) = \begin{pmatrix} 2u & 0 \\ 0 & 1 \end{pmatrix}, \qquad J_2F(u,v) = \sqrt{\det(DF^T DF)} = |\det DF| = 2|u|.$$

The critical set is $\{0\} \times [0,1]$, where the rank drops to one. Its image is the left edge of Q; both sets have $\mathcal{L}^2$-measure zero. We do not need to define an inverse derivative there.

3. Compare image area with the parameter integral. The image area is

$$\mathcal{H}^2(F(A)) = \mathcal{L}^2(Q) = 1.$$

Direct integration, justified by Tonelli for the nonnegative Borel integrand, gives

$$\int_A J_2F \, d\mathcal{L}^2 = \int_{-1}^{1} \int_0^1 2|u| \, dv \, du = 4 \int_0^1 u \, du = 2.$$

The area formula explains the factor:

$$\int_A J_2F \, d\mathcal{L}^2 = \int_{\mathbb{R}^2} N(F, A, z) \, d\mathcal{H}^2(z) = 2\mathcal{L}^2(Q).$$

The extra unit is an entire second sheet, not an error caused by a negligible boundary.

4. Keep the fibre sum in the weighted formula. Write the source weight as $h \circ F$, whose value at (u, v) is $u^2 + v$. The weighted area formula gives

$$\begin{aligned} \int_A (h \circ F) J_2F \, d\mathcal{L}^2 &= \int_{\mathbb{R}^2} \sum_{(u,v) \in A \cap F^{-1}(\{z\})} h(F(u,v)) \, d\mathcal{H}^2(z) \\ &= \int_Q h(z) N(F, A, z) \, d\mathcal{L}^2(z) \\ &= 2 \int_0^1 \int_0^1 (x + y) \, dy \, dx = 2. \end{aligned}$$

An independent direct calculation is

$$\begin{aligned} \int_{-1}^{1} \int_0^1 (u^2 + v) 2|u| \, dv \, du &= 4 \int_0^1 \left(u^3 + \frac{u}{2} \right) du \\ &= 4 \left(\frac{1}{4} + \frac{1}{4} \right) = 2. \end{aligned}$$

Thus a substitution that simply writes this parameter integral as $\int_Q h$ loses multiplicity.

5. Count the image once. Restrict F to $A_+ = [0,1] \times [0,1]$. This restriction is injective, including its boundary. Its Jacobian may vanish at $u = 0$; injectivity and Lipschitz regularity are still enough for the area formula. It follows that

$$\int_Q h \, d\mathcal{H}^2 = \int_0^1 \int_0^1 (u^2 + v) 2u \, dv \, du = 1.$$

Alternatively, divide the original parameter integral by two: $N(F, A, F(u,v)) = 2$ away from $u = 0$, and that exceptional line is null. For this particular map,

$$\int_Q h \, d\mathcal{H}^2 = \frac{1}{2} \int_A (h \circ F) J_2 F \, d\mathcal{L}^2 = 1.$$

A constant divisor works here because the multiplicity is constant almost everywhere on the image. Variable multiplicity requires the actual fibre count.

14.4 Why the theorem applies

The area formula applies to a Lipschitz map on a Lebesgue-measurable domain, with nonnegative measurable weights. Its target integral contains the sum over all preimages in that domain. Those hypotheses hold here, so absolute integrability need not be assumed before obtaining the nonnegative identity. Finiteness then follows from the explicit computations. For a signed weight one must separately check integrability, for example $\int_A |h \circ F| J_2 F < \infty$, before subtracting positive and negative parts.

14.5 Pitfalls

- The unsigned Jacobian is $2|u|$, not $2u$. Indeed $\int_A \det DF \, d\mathcal{L}^2 = 0$ by oddness: opposite orientations cancel. Scalar Hausdorff area is positive and does not cancel.
- Scalar Hausdorff integration supplies no normal vector. Vector flux requires an oriented normal and a dot product, in addition to the surface measure.
- Noninjectivity cannot be dismissed by deleting only the critical line: the two regular sheets still have the same image.
- The inverse branch $x \mapsto \sqrt{x}$ has an unbounded derivative near zero. That does not invalidate this forward Lipschitz area calculation.

14.6 Reusable technique

Find the fibres before changing variables. Compute $J_k F$ using $\sqrt{\det(DF^T DF)}$; keep the explicit preimage sum; then either choose an injective sheet or correct for verified multiplicity. Tags: `area-formula`, `multiplicity`, `fold`, `noninjective-parametrization`, `unsigned-jacobian`, `orientation`.

14.7 Sources

- Leon Simon, *Introduction to Geometric Measure Theory*, Chapter 2, §3, Theorem 3.3 and Remark 3.4(2), printed pp. 57–58; source ID `simon-gmt`. The same-dimensional specialization used here has target measure $\mathcal{H}^2 = \mathcal{L}^2$. The worked fold and calculations above are original to this collection.

15 E12 — Coarea for squared distance, including the critical value

15.1 Problem

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\}$ and define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) = x^2 + y^2$. For a nonnegative Borel function $\phi : [0, 1] \to [0, \infty]$, prove

$$\int_D \phi(g(x, y))\, d\mathcal{L}^2(x, y) = \pi \int_0^1 \phi(t)\, dt.$$

Use coarea, making the treatment of $g^{-1}(\{0\})$ explicit. Also compute $\int_D |\nabla g|\, d\mathcal{L}^2$ and determine exactly when the radial singularity $(x^2 + y^2)^a$ is integrable.

15.2 Data and hypotheses

Planar integration uses completed Lebesgue measure $\mathcal{L}^2$. The level sets carry normalized $\mathcal{H}^1$, so their measure is ordinary arc length. In general our normalization uses the covering cost $\alpha_k(\operatorname{diam} U)^k/2^k$, with α_k the unit k-ball volume; it gives $\mathcal{H}^k = \mathcal{L}^k$ on a k-plane and $\mathcal{H}^0$ equal to counting measure.

The map g is smooth on $\mathbb{R}^2$, the disk is Borel, and its one-dimensional coarea Jacobian is

$$J_1 g(x, y) = |\nabla g(x, y)| = 2\sqrt{x^2 + y^2}.$$

The weight ϕ may be unbounded, and either side of the requested formula may be $+\infty$. No signed cancellation is permitted in this extended nonnegative statement. A value assigned to a function at the single point $(0, 0)$ will not change its $\mathcal{L}^2$ integral.

15.3 Solution

1. Describe the levels and the critical set. Put $r = \sqrt{x^2 + y^2}$. For $0 < t \le 1$ the level in D is the circle of radius $\sqrt{t}$:

$$D \cap g^{-1}(\{t\}) = \{(\sqrt{t}\cos\theta, \sqrt{t}\sin\theta) : 0 \le \theta < 2\pi\}.$$

Its normalized $\mathcal{H}^1$ measure is $2\pi\sqrt{t}$. One can check this by the injective arc parametrization on $[0, 2\pi)$: the speed is $\sqrt{t}$, so its length is $\int_0^{2\pi} \sqrt{t}\, d\theta$. At $t = 0$ the level is the singleton $\{(0, 0)\}$, whose $\mathcal{H}^1$ measure is zero. Outside $[0, 1]$ the level in D is empty.

The only critical point is $(0, 0)$ because $\nabla g = (2x, 2y)$. The only critical value is therefore 0. Here this is a direct calculation, requiring no appeal to Sard's theorem. The regular levels are circles; the exceptional level is a point. Treating it separately avoids a division by zero.

2. First apply coarea without dividing. The weighted coarea formula for this smooth scalar map is

$$\int_{\mathbb{R}^2} w(z)|\nabla g(z)|\, d\mathcal{L}^2(z) = \int_{\mathbb{R}} \left(\int_{g^{-1}(\{t\})} w(z)\, d\mathcal{H}^1(z) \right) dt$$

for nonnegative Borel w, with the usual convention that the product is zero where the Jacobian is zero. Taking $w = \mathbf{1}_D$ gives

$$\int_D |\nabla g|\, d\mathcal{L}^2 = \int_0^1 2\pi\sqrt{t}\, dt = \frac{4\pi}{3}.$$

The value of the inner integral at $t = 0$ is exactly zero. The value at the endpoint $t = 1$ does not affect its one-dimensional Lebesgue integral.

3. Divide only on the regular part. Define a Borel function $w : \mathbb{R}^2 \to [0, \infty]$ by

$$w(z) = \begin{cases} \dfrac{\phi(g(z))}{2|z|}, & z \in D \setminus \{0\}, \\ 0, & z = 0 \text{ or } z \notin D. \end{cases}$$

On $D \setminus \{0\}$ one has $wJ_1 g = \phi \circ g$. At zero the two functions may have different values, but this is a $\mathcal{L}^2$-null set. Consequently their nonnegative Lebesgue integrals agree, even if $\phi(0) = +\infty$.

For $0 < t \leq 1$, w is constant on the level circle, with value $\phi(t)/(2\sqrt{t})$. Hence the inner level integral is

$$\int_{g^{-1}(\{t\})} w \, d\mathcal{H}^1 = \frac{\phi(t)}{2\sqrt{t}} \, 2\pi\sqrt{t} = \pi\phi(t).$$

At $t = 0$, the definition $w(0) = 0$ makes the inner integral zero. Thus coarea proves

$$\int_D \phi \circ g \, d\mathcal{L}^2 = \int_{(0,1]} \pi\phi(t) \, dt = \pi \int_0^1 \phi(t) \, dt.$$

Only the integrals ignore the endpoints; no claim that $1/|\nabla g|$ is defined at every point has been made.

4. Read off useful consequences. For a real exponent a, assign any finite value at the origin to the function $z \mapsto |z|^{2a}$ if its literal formula is undefined there. On $(0, 1]$ use $\phi(t) = t^a$, and assign $\phi(0) = 0$. The preceding identity gives

$$\int_D |z|^{2a} \, d\mathcal{L}^2(z) = \begin{cases} \dfrac{\pi}{a+1}, & a > -1, \\ +\infty, & a \leq -1. \end{cases}$$

In particular $|z|^{-1}$ is integrable on the disk, with integral 2π, whereas $|z|^{-2}$ is not. The divergence occurs throughout arbitrarily small neighborhoods of zero; changing the value at zero cannot repair it.

For $0 \leq a_0 < b_0 \leq 1$, taking $\phi = \mathbf{1}_{(a_0, b_0)}$ gives the annular area $\pi(b_0 - a_0)$ for $a_0 < |z|^2 < b_0$. Equivalently, if a random point is uniform in D with probability measure $\pi^{-1}\mathcal{L}^2|_D$, its squared radius is uniform on $[0, 1]$. These are both consequences of the same pushforward identity, not additional assumptions about polar coordinates.

15.4 Why the theorem applies

The locally Lipschitz weighted coarea theorem applies to g on $\mathbb{R}^2$, with the Borel weights above. Its Jacobian is the norm of the gradient because the target dimension is one. This polynomial is smooth and locally Lipschitz; global Lipschitz continuity on all of $\mathbb{R}^2$ is unnecessary. If using a formulation requiring a globally Lipschitz map, use $G(z) = \min\{|z|^2, 4\}$. The function $r \mapsto \min\{r^2, 4\}$ is 4-Lipschitz on $[0, \infty)$, and $z \mapsto |z|$ is 1-Lipschitz, so G is globally 4-Lipschitz and agrees with g on a neighborhood of D.

For signed Borel ϕ with $\int_0^1 |\phi(t)| \, dt < \infty$, first apply the nonnegative identity to $|\phi|$. This proves absolute integrability on D; applying the formula to ϕ^+ and ϕ^- then gives the signed identity. If both sign parts have infinite integral, their difference is undefined.

15.5 Pitfalls

- Coarea gives a factor $|\nabla g| = 2r$, not its reciprocal. A reciprocal appears only because we deliberately choose a compensating weight on the regular part.
- A critical level need not be a regular curve. Its contribution must be checked, rather than parametrized as though its dimension were unchanged.
- The critical point being null does not by itself establish integrability of a singular weight near it; the power computation supplies the needed neighborhood estimate.
- These are scalar integrals over levels. No orientation or vector flux occurs.

15.6 Reusable technique

For an integral depending on a scalar map g, compute the geometry of $g^{-1}(\{t\})$ and the coarea Jacobian, isolate the critical set, and choose a compensating weight only where the Jacobian is nonzero. Tags: `coarea`, `level-sets`, `radial-integral`, `critical-value`, `singular-weight`, `pushforward-measure`.

15.7 Sources

- Leon Simon, *Introduction to Geometric Measure Theory*, Chapter 2, §7, Theorem 7.3, Remark 7.4(1), and Theorem 7.5, printed pp. 76, 80; source ID `simon-gmt`. The weighted coarea identity is the theorem used; the critical point and all integrals in this example are computed directly. Chapter 2, §3, Example 3.5(1) connects normalized $\mathcal{H}^1$ with arc length.

16 E13 — A singular map loses all its increase under differentiation

16.1 Problem

Construct a continuous nondecreasing map $C : [0,1] \to [0,1]$ with $C(0) = 0$, $C(1) = 1$, and $C' = 0$ almost everywhere. Compare

$$\int_0^1 |C'(x)|\,dx \quad \text{with} \quad \int_{\mathbb{R}} \mathcal{H}^0(C^{-1}(\{y\}))\,dy.$$

Explain exactly why the Lipschitz area formula and the fundamental theorem for absolutely continuous functions cannot be applied to C. Use the construction to exhibit concentration that prevents exchanging a derivative limit with an integral, even though the functions themselves converge uniformly.

16.2 Data and hypotheses

The source and target use completed one-dimensional Lebesgue measure $\mathcal{L}^1$. Normalized $\mathcal{H}^1$ equals $\mathcal{L}^1$ on a line; the covering normalization in dimension $k \geq 1$ is $\alpha_k(\operatorname{diam} U)^k/2^k$. The measure $\mathcal{H}^0$ counts points, so an infinite fibre has value $+\infty$.

Derivatives will be used as almost-everywhere defined functions. Any undefined values on an explicitly identified null set may be assigned zero to obtain a measurable representative. This convention does not assert differentiability there. All finite-stage approximations below are piecewise linear and absolutely continuous; their limit will fail absolute continuity.

16.3 Solution

1. Build the set and the approximations. Let $K_0 = [0,1]$. Obtain K_{n+1} from K_n by removing the open middle third of each component interval. The set K_n has 2^n disjoint closed component intervals, each of length 3^{-n}. Thus

$$\mathcal{L}^1(K_n) = 2^n 3^{-n} = \left(\frac{2}{3}\right)^n.$$

The compact Cantor set $K = \bigcap_{n=0}^{\infty} K_n$ is null by continuity of finite measure from above. It therefore has empty interior, because every nondegenerate interval has positive Lebesgue measure.

Define C_n to be the continuous nondecreasing function that is constant on each gap removed by stage n, rises linearly by 2^{-n} on each component interval of K_n, and satisfies $C_n(0) = 0$. These rules uniquely determine it, with $C_n(1) = 1$. For example $C_0(x) = x$, while C_1 rises from 0 to $1/2$ on $[0, 1/3]$, is constant $1/2$ on $[1/3, 2/3]$, and rises to 1 on $[2/3, 1]$.

All later approximations retain the endpoint values of every stage-n component interval. Within such an interval both C_m and C_n, for $m \geq n$, remain between endpoint values separated by 2^{-n}; on previously removed gaps they agree. Therefore

$$\sup_{x \in [0,1]} |C_m(x) - C_n(x)| \leq 2^{-n} \qquad (m \geq n).$$

The sequence converges uniformly to a continuous nondecreasing function C. The endpoint values persist, so $C(0) = 0$, $C(1) = 1$, and the intermediate value theorem gives $C([0,1]) = [0,1]$. This construction avoids all ambiguity of ternary expansions at interval endpoints.

2. Compute the almost-everywhere derivative. Every $x \in [0,1] \setminus K$ lies in an open gap removed at a finite stage. All sufficiently late C_n, and hence C, are constant on that gap. Thus $C'(x) = 0$ at every such x. Since $\mathcal{L}^1(K) = 0$, $C' = 0$ almost everywhere. Assigning zero on K yields the zero function as a measurable representative, and therefore

$$\int_0^1 |C'| \, d\mathcal{L}^1 = 0, \qquad C(1) - C(0) = 1.$$

We do not claim that C' exists at every point of K.

3. Compute the multiplicity on the target. Let P be the set of constant values of C on its deleted open gaps. There are countably many gaps, so P is countable; these plateau values are dyadic rationals. If two points $a < b$ satisfy $C(a) = C(b) = y$, monotonicity forces C to be constant on $[a, b]$. Because K has empty interior, (a, b) contains part of a deleted gap, so $y \in P$. Consequently every $y \in [0,1] \setminus P$ has exactly one preimage, using surjectivity for existence. Every $y \in P$ has infinitely many preimages, including an entire gap. Thus the counting function N has values

$$N(C, [0,1], y) = \begin{cases} 1, & y \in [0,1] \setminus P, \\ +\infty, & y \in P, \\ 0, & y \notin [0,1]. \end{cases}$$

It is measurable, and the infinite values on the countable null set P do not change its nonnegative Lebesgue integral. Hence

$$\int_{\mathbb{R}} N(C, [0,1], y) \, dy = 1 \neq 0 = \int_0^1 |C'| \, dx.$$

This is a failed extension of the Lipschitz area identity to a continuous, almost-everywhere differentiable map.

4. Identify the exact regularity failure. On the leftmost stage-n interval, $C(3^{-n}) - C(0) = 2^{-n}$. The corresponding difference quotient is $(3/2)^n$, so no finite Lipschitz constant exists.

Absolute continuity fails directly as well. For the 2^n basic intervals $[a_{n,j}, b_{n,j}]$ at stage n,

$$\sum_{j=1}^{2^n}(b_{n,j} - a_{n,j}) = (2/3)^n \longrightarrow 0, \qquad \sum_{j=1}^{2^n}|C(b_{n,j}) - C(a_{n,j})| = \sum_{j=1}^{2^n} 2^{-n} = 1.$$

Taking $\varepsilon = 1/2$ contradicts the defining absolute-continuity condition for every proposed $\delta > 0$.

There is also an image-of-null-set obstruction. Every plateau value has a preimage at a gap endpoint belonging to K, while any preimage outside K lies in a gap. Together with surjectivity this proves $C(K) = [0, 1]$. Thus this map sends a null set to a set of measure one. In contrast, a Lipschitz map on a line sends null sets to null sets: cover the source null set by intervals of arbitrarily small total length, then cover their images by intervals whose lengths are at most the Lipschitz constant times those lengths. Discarding a source null set is therefore unsafe for this singular map.

5. See concentration in the derivatives. At all but finitely many breakpoints,

$$C_n'(x) = \left(\frac{3}{2}\right)^n \mathbf{1}_{K_n}(x).$$

Choose the displayed right side as an almost-everywhere representative, denoted q_n. Then

$$\int_0^1 q_n(x)\,dx = \left(\frac{3}{2}\right)^n \left(\frac{2}{3}\right)^n = 1, \qquad q_n \longrightarrow 0 \quad \mathcal{L}^1\text{-almost everywhere.}$$

Thus $\int \lim q_n = 0$ while $\lim \int q_n = 1$. The mass stays inside a bounded interval and concentrates on the shrinking sets K_n containing the null set K; it does not escape to infinity.

There is no common integrable dominator. If $q_n \le G$ almost everywhere for $G \in L^1$, then $\int_{K_n} |G| \ge 1$ for every n. Dominated convergence applied to $|G|\mathbf{1}_{K_n} \to 0$ almost everywhere would instead give $\int_{K_n} |G| \to 0$, a contradiction. The sequence also fails uniform integrability, since $\mathcal{L}^1(K_n) \to 0$ but $\int_{K_n} q_n = 1$.

16.4 Why the theorem applies

The continuity-of-measure, uniform-limit, and nonnegative integration arguments used in the construction have their stated hypotheses. The area formula is used only as a comparison theorem: its Lipschitz hypothesis fails for C, so its equality is not asserted. The absolute-continuity fundamental theorem similarly does not apply to C.

A useful positive replacement is the following. If F_n are absolutely continuous on $[0, 1]$, $F_n(0) \to a$, $F_n' \to q$ almost everywhere, and $|F_n'| \le G \in L^1$ almost everywhere, then dominated convergence gives

$$\sup_{x\in[0,1]} \left| F_n(x) - a - \int_0^x q(t)\,dt \right| \le |F_n(0) - a| + \int_0^1 |F_n' - q|\,dt \longrightarrow 0.$$

Here $q \in L^1$ because $|q| \le G$ almost everywhere, and the limit is absolutely continuous with derivative q almost everywhere. This is a precise sufficient condition for retaining derivatives under the limit; the Cantor approximations violate the domination condition.

16.5 Pitfalls

- Uniform convergence of functions does not supply domination of their derivatives.
- An almost-everywhere zero derivative alone does not make a continuous function constant. Absolute continuity restores that conclusion.
- An infinite multiplicity at countably many target values is harmless for this target integral; a null source set with a full-measure image is a different issue.
- This does not contradict a smooth Sard theorem: C is not a smooth map, and the exceptional domain set cannot be removed while preserving its image measure.

16.6 Reusable technique

Track where derivative mass lives, not just the pointwise limit. Check absolute continuity or the actual regularity required by area/change-of-variables formulas; test whether null source sets can have large images. Tags: `singular-map`, `cantor-function`, `absolute-continuity`, `area-formula-failure`, `concentration`, `dominated-convergence-failure`, `uniform-integrability`, `derivative-limit`.

16.7 Sources

- John K. Hunter, *Measure Theory*, Example 2.14 (Cantor-set nullity), Example 2.37 (singular Cantor measure), and the opening of Chapter 6, printed pp. 19, 31, 63; source ID `hunter-measure`. These give background on the standard set and the absolute-continuity obstruction; the construction and calculations above are supplied explicitly.
- Leon Simon, *Introduction to Geometric Measure Theory*, Chapter 2, §3, Theorem 3.3 and Remark 3.4(2), printed pp. 57–58; source ID `simon-gmt`. The Lipschitz area theorem is the precise formula whose unjustified extension is disproved here.

17 E14 — Lost integral mass: concentration, escape, and dilution

17.1 Problem

Compare the limits and integrals of these measurable functions:

$$f_n(x) = n\mathbf{1}_{(0,1/n)}(x) \quad (x \in (0,1)), \qquad h_n(x) = \mathbf{1}_{(n,n+1)}(x) \quad (x \in \mathbb{R}).$$

Explain why dominated convergence fails. Identify sufficient replacement hypotheses, keeping finite and infinite measure spaces distinct.

17.2 Data and hypotheses

All measures are one-dimensional Lebesgue measure. The symbol f_n denotes a function; $f_n(x)$ denotes its value. All three sequences used below consist of nonnegative Borel functions. Convergence in measure on a measure space means $\mu(\{|u_n - u| > \varepsilon\}) \to 0$ for each $\varepsilon > 0$. "Locally in measure" on $\mathbb{R}$ means this on every bounded interval.

We use **value-tail uniform integrability** to mean

$$\lim_{K\to\infty} \sup_n \int_{\{|u_n|>K\}} |u_n|\, d\mu = 0.$$

On finite measure spaces this is the usual uniform-integrability criterion. On infinite measure spaces some definitions additionally require spatial tightness; here that condition will always be stated separately.

17.3 Solution

1. Concentration on a shrinking set. For each fixed $x \in (0,1)$, eventually $n > 1/x$, so $f_n(x) = 0$. Nevertheless

$$\int_0^1 f_n(x)\,dx = n\lambda^1((0,1/n)) = 1.$$

Thus $f_n \to 0$ pointwise everywhere on its domain, while $\int f_n \not\to \int 0$. It also converges in measure since, for $n > \varepsilon$,

$$\lambda^1(\{f_n > \varepsilon\}) = 1/n \longrightarrow 0.$$

There is no common integrable majorant G: if $G \geq f_n$ almost everywhere for all n, take a common full-measure set by removing the countable union of exceptional sets. Then $\int_0^{1/n} G \geq 1$ for every n, contradicting absolute continuity of the integral of an L^1 function as $1/n \to 0$. Its value-tail integrals also show precisely what fails: for any K, choose an integer $n > K$ to obtain

$$\int_{\{f_n > K\}} f_n\,dx = 1.$$

2. Escape to infinity without large values. For each $x \in \mathbb{R}$, $h_n(x) = 0$ eventually, but $\int_{\mathbb{R}} h_n = 1$. The functions are bounded by 1, which is not integrable on $\mathbb{R}$. A common L^1 majorant is impossible because the intervals $(n, n+1)$ are disjoint and

$$\int_{\mathbb{R}} G\,dx \geq \sum_{n=1}^{\infty} \int_n^{n+1} G\,dx \geq \sum_{n=1}^{\infty} 1 = \infty.$$

Here value-tail uniform integrability holds: for $K \geq 1$ every tail integral is zero. What fails is **spatial tightness**:

$$\lim_{R\to\infty} \sup_n \int_{\{|x|>R\}} |h_n(x)|\,dx = 1,$$

not zero. Also $h_n \to 0$ locally in measure but not in measure on all of $\mathbb{R}$, since $\lambda^1(\{h_n > 1/2\}) = 1$.

3. Even global convergence in measure is insufficient on an infinite space. Set $r_n(x) = n^{-1}\mathbf{1}_{(0,n)}(x)$ on $\mathbb{R}$. Then $r_n \to 0$ pointwise and uniformly, hence in measure, and $\int r_n = 1$. Its values are at most 1, so it is value-tail uniformly integrable. But for fixed $R > 0$,

$$\int_{\{|x|>R\}} r_n\,dx = \max(1 - R/n, 0),$$

whose supremum over n is 1. Mass spreads out at smaller heights. This is why a finite-measure version of a theorem cannot silently be applied to $\mathbb{R}$.

17.4 Why the theorem applies

Useful sufficient conditions are:

- On any measure space, if $u_n \to u$ almost everywhere and $|u_n| \leq G$ almost everywhere for one $G \in L^1$, dominated convergence gives $\|u_n - u\|_1 \to 0$ and therefore convergence of integrals.

- On a **finite** measure space, if u_n and u are real measurable functions, $u_n \in L^1$, $u_n \to u$ in measure, and the sequence is value-tail uniformly integrable, then $u \in L^1$ and $\|u_n - u\|_1 \to 0$.
- On $\mathbb{R}$, for real measurable u_n, u with $u_n \in L^1$, the same conclusion follows from local convergence in measure, value-tail uniform integrability, and spatial tightness $\lim_{R\to\infty} \sup_n \int_{|x|>R} |u_n| = 0$.

For completeness, the finite-measure assertion follows by truncation: let $T_K(s) = \max(-K, \min(s, K))$ for real functions. On a finite space, convergence in measure gives

$$\int |T_K(u_n) - T_K(u)| \le \varepsilon\mu(X) + 2K\mu(\{|u_n - u| > \varepsilon\}) \longrightarrow \text{at most } \varepsilon\mu(X).$$

Uniform integrability controls $\|u_n - T_K(u_n)\|_1$ uniformly. A subsequence converges almost everywhere (choose error probabilities summable); Fatou bounds the corresponding tail for u by the same uniform tail bound. Let $K \to \infty$ and then $\varepsilon \to 0$. On $\mathbb{R}$, apply this finite-space argument on $[-R, R]$ and control its complement by spatial tightness. To bound the tail of u, choose a diagonal subsequence converging almost everywhere on each $[-m, m]$: at its jth step make the measure of the error set $\{|u_{n_j} - u| > 1/j\} \cap [-j, j]$ less than 2^{-j}. The summable-error argument gives almost-everywhere convergence on all of $\mathbb{R}$, and Fatou then bounds $\int_{|x|>R} |u|$ by the uniform tail bound. These are analytic hypotheses, not conclusions inferred from numerical samples.

17.5 Pitfalls

Each $\int u_n$ being finite, or their values being uniformly bounded, is insufficient. Uniform convergence alone does not control integrals over an infinite-measure domain. No example contradicts Fatou: it only asserts $0 = \int \liminf u_n \le \liminf \int u_n = 1$ here.

17.6 Reusable technique

Before moving a limit inside an integral, inspect large values on small sets and mass outside large bounded sets separately. Tags: `dominated-convergence`, `uniform-integrability`, `concentration`, `escape-to-infinity`, `tightness`, `counterexample`.

17.7 Sources

- John K. Hunter, *Measure Theory*, Examples 4.18–4.20 (the standard concentration, dilution and escape families), Theorem 4.24 (dominated convergence), the following discussion of absolute continuity and uniform integrability, and Chapter 4's Fatou lemma. The computations and additional uniform-integrability/tightness analysis here are worked through explicitly. Source ID: `hunter-measure`.
- The uniform-integrability implications used here are proved in the paragraph above; the source is not being cited as containing this exact formulation or atlas.

18 E15 — A parameter derivative misses a concentrating boundary layer

18.1 Problem

On $x \in (0, 1)$, for $t \ge 0$, define

$$f(t, x) = \begin{cases} \dfrac{t^2}{x^2 + t^2}, & t > 0, \\ 0, & t = 0, \end{cases} \qquad I(t) = \int_0^1 f(t, x)\, dx.$$

Compare $I'_+(0)$ with $\int_0^1 \partial_t^+ f(0,x)\,dx$. Locate the precise missing hypothesis for differentiation under the integral sign.

18.2 Data and hypotheses

The measure is Lebesgue measure λ^1 on $(0,1)$. For every t, $x \mapsto f(t,x)$ is Borel and satisfies $0 \le f(t,x) \le 1$. For each fixed $x > 0$, $t \mapsto f(t,x)$ has right derivative at zero. We ask about a derivative of the **integral-valued function** I, not about a derivative of one fixed value $f(t,x)$.

18.3 Solution

1. Evaluate the integral before differentiating. For $t > 0$, substitute $x = tu$:

$$I(t) = t\int_0^{1/t} \frac{1}{1+u^2}\,du = t\arctan(1/t), \qquad I(0) = 0.$$

It follows that

$$I'_+(0) = \lim_{t\downarrow 0} \frac{I(t)-I(0)}{t} = \lim_{t\downarrow 0} \arctan(1/t) = \frac{\pi}{2}.$$

2. Differentiate each fixed section. For every fixed $x > 0$,

$$\partial_t^+ f(0,x) = \lim_{t\downarrow 0} \frac{f(t,x)-f(0,x)}{t} = \lim_{t\downarrow 0} \frac{t}{x^2+t^2} = 0.$$

Thus $\int_0^1 \partial_t^+ f(0,x)\,dx = 0$, which differs from $I'_+(0)$.

3. Find the boundary layer. The difference-quotient functions are

$$q_t(x) = \frac{t}{x^2+t^2} \ge 0.$$

They tend to zero at each fixed $x > 0$, but

$$\int_0^1 q_t(x)\,dx = \arctan(1/t) \longrightarrow \pi/2.$$

For $0 < t \le 1$ the interval $(0,t)$ already carries a fixed amount of mass:

$$\int_0^t q_t(x)\,dx = \int_0^1 \frac{du}{1+u^2} = \pi/4.$$

This proves that the family of quotients has no common $L^1(0,1)$ majorant on any neighborhood $0 < t < \delta$: otherwise absolute continuity of its integral would force the displayed integrals to tend to zero.

The derivative family also has no local common integrable majorant. For $t > 0$,

$$\partial_t f(t,x) = \frac{2tx^2}{(x^2+t^2)^2}.$$

To avoid any uncountable exceptional-set issue, suppose $G \in L^1$ bounds this derivative for almost every x for **each** $t \in (0,\delta)$. Take $t_j \downarrow 0$ with $2t_j < 1$. On $(t_j, 2t_j)$, put $x = t_j u$ to obtain

$$\int_{t_j}^{2t_j} \partial_t f(t_j,x)\,dx = \int_1^2 \frac{2u^2}{(1+u^2)^2}\,du > 0,$$

independent of j. Since G dominates the integrand at these countably many parameters, absolute continuity again gives a contradiction. This even rules out bounds whose null exceptional set was initially allowed to depend on the parameter.

18.4 Why the theorem applies

Ordinary dominated convergence **does** give $I(t) \to 0$: $f(t,x) \to 0$ and $|f(t,x)| \le 1 \in L^1(0,1)$. It says nothing about the derivative unless the quotients are controlled.

A precise sufficient differentiation rule is as follows. Let $(X, \mathcal{A}, \mu)$ be a measure space and J an open real interval containing t_0. Assume $F(t,\cdot)$ is measurable for every $t \in J$, $F(t_0,\cdot) \in L^1(\mu)$, and there is a **common** full-measure set on which $t \mapsto F(t,x)$ is differentiable throughout J and

$$|\partial_t F(t,x)| \le G(x) \quad (t \in J), \qquad G \in L^1(\mu).$$

Then $F(t,\cdot) \in L^1$ near t_0 and

$$\frac{d}{dt}\Big|_{t=t_0} \int_X F(t,x)\, d\mu(x) = \int_X \partial_t F(t_0,x)\, d\mu(x).$$

Indeed the mean value theorem bounds the difference quotients by G, and dominated convergence proves the identity along every sequence of nonzero increments tending to zero. The same proof works for a right derivative at a left endpoint, using a half-interval and right differentiability at the endpoint. In this example the required bound fails exactly where x and t approach zero together.

18.5 Pitfalls

- An integrable bound on f is not an integrable bound on its parameter derivative or difference quotients.
- Differentiability at every fixed x does not imply differentiation can pass through integration.
- A related warning for a different exchange is $v_n(x) = \sin(nx)/n$ on $[-1,1]$: $v_n \to 0$ uniformly, but $v_n'(0) = 1$ for all n, whereas the limit function has derivative 0. Uniform convergence of functions alone does not justify exchanging a limit and a derivative.

18.6 Reusable technique

Write the actual difference quotients and search for a common integrable bound. When the quotient concentrates, integrate it over a parameter-dependent shrinking interval to prove that no such bound exists. Tags: `differentiate-under-integral`, `boundary-layer`, `concentration`, `dominated-convergence`, `counterexample`.

18.7 Sources

- John K. Hunter, *Measure Theory*, Theorem 4.24 (the DCT used in the proof). Source ID: `hunter-measure`.
- Keith Conrad, *Differentiating under the integral sign*, Section 12 and Theorem 12.3, for classical hypotheses and a warning example. The measurable sufficient rule here is explicitly proved above; the present boundary-layer example is independently worked. Source ID: `conrad-diff`.

19 E16 — An exponential kernel retrieves a boundary value

19.1 Problem

Let $g : [0,\infty) \to \mathbb{R}$ be bounded and Borel measurable, and suppose it is right-continuous at 0. Prove

$$\lim_{n\to\infty} \int_0^\infty n e^{-nx} g(x)\, dx = g(0).$$

Give a direct error estimate, and identify why applying dominated convergence in the original variable would be unsafe.

19.2 Data and hypotheses

The measure is one-dimensional Lebesgue measure on $[0,\infty)$. Choose $M \geq 0$ with $|g(x)| \leq M$ for all $x \geq 0$. The kernel function is $k_n : x \mapsto ne^{-nx}$; its value at x is $k_n(x)$. It satisfies

$$k_n \geq 0, \qquad \int_0^\infty k_n(x)\,dx = 1, \qquad \int_\delta^\infty k_n(x)\,dx = e^{-n\delta} \quad (\delta > 0).$$

Every displayed integral of $k_n g$ is absolutely convergent because $\int |k_n g| \leq M$. The value $g(0)$ is recoverable under the stated continuity assumption; Lebesgue integration by itself ignores a single point.

19.3 Solution

1. Rescale the moving kernel. The substitution $u = nx$, with $dx = du/n$, gives

$$\int_0^\infty ne^{-nx} g(x)\,dx = \int_0^\infty e^{-u} g(u/n)\,du.$$

For each fixed $u \geq 0$, right-continuity gives $g(u/n) \to g(0)$. Moreover

$$|e^{-u} g(u/n)| \leq Me^{-u}, \qquad \int_0^\infty Me^{-u}\,du = M < \infty.$$

Dominated convergence now applies on the fixed u-domain and gives

$$\lim_n \int_0^\infty e^{-u} g(u/n)\,du = g(0) \int_0^\infty e^{-u}\,du = g(0).$$

2. Obtain a direct epsilon and tail proof. For any $\varepsilon > 0$, choose $\delta > 0$ such that $|g(x) - g(0)| < \varepsilon$ when $0 \leq x < \delta$. Normalization of k_n yields

$$\begin{aligned}\left| \int_0^\infty k_n(x) g(x)\,dx - g(0) \right| &\leq \int_0^\delta k_n(x) |g(x) - g(0)|\,dx + \int_\delta^\infty k_n(x) |g(x) - g(0)|\,dx \\ &\leq \varepsilon + 2Me^{-n\delta}.\end{aligned}$$

For this fixed δ, the tail tends to zero. Then let $\varepsilon \downarrow 0$.

3. A useful convergence rate. If, additionally, $|g(x) - g(0)| \leq Lx$ for every $x \geq 0$ (in particular if g is globally L-Lipschitz), then

$$\left| \int_0^\infty k_n(x) g(x)\,dx - g(0) \right| \leq L \int_0^\infty ne^{-nx} x\,dx = \frac{L}{n} \int_0^\infty ue^{-u}\,du = \frac{L}{n}.$$

The last integral is 1 by integration by parts. If only a local Lipschitz bound holds for $0 \leq x < \delta$, the same split gives the valid bound $L/n + 2Me^{-n\delta}$.

19.4 Why the theorem applies

The rescaled integrands $u \mapsto e^{-u} g(u/n)$ are Borel, converge pointwise, and have the explicit common L^1 majorant Me^{-u}. These are precisely the dominated-convergence hypotheses. The substitution is a positive linear change of variables, so its Jacobian is $1/n$; it is valid for these absolutely integrable measurable functions.

In the original variable $k_n(x) \to 0$ for every $x > 0$, while $\int k_n = 1$. Hence no common L^1 function dominates all k_n. More directly, if $G \geq k_n$ almost everywhere for all n, then

$$\int_0^{1/n} G\,dx \geq \int_0^{1/n} ne^{-nx}\,dx = 1 - e^{-1},$$

contradicting absolute continuity of an L^1 integral. Rescaling resolves the concentration instead of incorrectly ignoring it. In measure language the probability measures $k_n(x)\,dx$ converge weakly against bounded continuous test functions on $[0, \infty)$ to the unit point mass at 0; they do not converge to the zero measure merely because their densities tend to zero almost everywhere.

19.5 Pitfalls

- Do not infer the limit integral is zero from $ne^{-nx} g(x) \to 0$ for $x > 0$: a common integrable bound has not been supplied.
- If right-continuity is removed, the answer need not equal the chosen value at zero. For $g(0) = 1$ and $g(x) = 0$ for $x > 0$, every integral is zero.
- Boundedness and continuity at zero supply different parts of the proof: boundedness controls the far tail, while continuity identifies the limit near the boundary.

19.6 Reusable technique

For a normalized concentrating kernel, either rescale to a fixed kernel and use DCT, or split into a small neighborhood and an exponentially small tail. Tags: `approximate-identity`, `dominated-convergence`, `rescaling`, `boundary-value`, `error-bound`, `laplace-kernel`.

19.7 Sources

- John K. Hunter, *Measure Theory*, Theorem 4.24 (dominated convergence) and Chapter 4's absolute continuity of the integral. Source ID: `hunter-measure`.
- The exponential-kernel calculation, estimates, and rate are proved directly above; no distributional formalism is required.

20 A small atlas of failed exchanges

This is a retrieval map to complete proofs, not a replacement for them. It gives outcomes; use PROBLEMS.md for the problem-first route.

Exchange or tempting inference	Example and failure mechanism	Positive sufficient condition
Integrate every section of a Lebesgue-measurable set	E03: a nonmeasurable set can sit in one exceptional section of a planar null set.	Borel measurability gives Borel sections everywhere; completed Lebesgue measurability gives measurable sections almost everywhere.
Swap signed iterated integrals	E04: both signed parts have infinite integral near the origin. E05: the signed matrix has infinite absolute sum.	Joint measurability and finite integral of the absolute value give Fubini; nonnegativity gives Tonelli with extended values.
Pass an almost-everywhere limit through an integral	E14: mass concentrates at a shrinking interval; its height grows.	DCT with one integrable majorant; on a finite measure space, convergence in measure plus uniform integrability gives convergence in L^1.
Treat bounded integral mass as sufficient on an infinite space	E14: a unit bump moves to infinity. Value-tail uniform integrability alone permits this.	Add spatial tightness to value-tail uniform integrability, together with convergence in measure on bounded sets (and an integrable limit). Precise versions are stated in E14.
Differentiate a parameter integral at an endpoint	E15: difference-quotient mass concentrates near zero; pointwise derivatives miss that mass.	Measurability, pointwise differentiability almost everywhere, one integrable local bound on the parameter derivative, and one integrable parameter slice. See the exact interval and endpoint conditions in E15.
Differentiate a uniform limit of differentiable functions	E15 related example: $\sin(nx)/n$ tends uniformly to zero while its derivative is 1 at zero for every n.	On a compact interval, uniform convergence of the derivatives and convergence of the function values at one point imply a differentiable limit with the expected derivative. The mean value theorem proves this for differentiable functions; for C^1 functions one can also use the fundamental theorem of calculus.
Replace geometric area by parameter area	E08/E11: repeated or folded parametrizations count the image more than once.	Injectivity outside sets whose contributions are zero, or explicit multiplicity in the area formula.

Exchange or tempting inference	Example and failure mechanism	Positive sufficient condition
Divide by a coarea Jacobian everywhere	E12: the gradient vanishes at the origin.	Restrict division to the positive-Jacobian set, then prove the zero set contributes zero or account for its contribution separately.
Use the area formula for a merely continuous monotone map	E13: the Cantor staircase has derivative zero almost everywhere but maps onto an interval.	Lipschitz maps satisfy the area formula; in one dimension absolute continuity gives the fundamental theorem and the relevant change-of-variable identities.

Concentration is also a tool: E16 uses an approximate identity. There the integration variable is rescaled and a fixed integrable majorant becomes available. The conclusion is a boundary evaluation, with all its hypotheses stated, rather than an exchange based solely on pointwise convergence in the original variable.

21 Sources and conventions audit

All links below were opened and checked on **2026-10-15**. These are author-hosted teaching texts, used for theorem statements and conventions. The worked examples in this directory are original expositions; the linked texts' proofs and problem collections are not reproduced. PDF page numbers below count the cover as page 1. A PDF viewer may display a different logical page label. Online files can change.

21.1 `hunter-measure`

John K. Hunter, *Measure Theory*, University of California, Davis, 2011. Author's PDF.

- Monotone convergence: Theorem 4.6, printed p.41, PDF p.45.
- Dominated convergence: Theorem 4.24, printed p.49, PDF p.53; a.e. versions are discussed in §4.6, printed p.50.
- Product-measurable sections: §5.1, Proposition 5.2, printed p.55, PDF p.59; function sections: Theorem 5.15, printed p.60.
- Fubini with absolute integrability: Theorem 5.18, printed p.61, PDF p.65.
- Completion and exceptional sections: Example 5.20 and Theorem 5.21, printed p.62, PDF p.66.

The completed Lebesgue product statement guarantees measurable and integrable sections almost everywhere. It does not guarantee measurability of every section. Theorem 5.15's sentence defining complex section integrals suppresses existence hypotheses; use the explicit integrability hypotheses of 5.18/5.21.

21.2 `axler-mira`

Sheldon Axler, *Measure, Integration & Real Analysis*, Graduate Texts in Mathematics 282, Springer, 2020; author's online revision dated 12 June 2026. Author's PDF.

- Tonelli: §5B, Theorem 5.28, printed p.129, PDF p.144.
- Nonnegative double sums: Corollary 5.31, printed p.131, PDF p.146.
- Fubini: Theorem 5.32, printed p.132, PDF p.147, with exceptional-parameter conventions immediately above and in the proof.

These product theorems assume sigma-finite factors. Tonelli accepts nonnegative product-measurable functions and permits the value infinity; Fubini requires a finite integral of the absolute value. This is an independent reference for the distinction used in the examples. For completed Lebesgue measurability rather than product measurability, also use `hunter-measure` §5.7.

21.3 `simon-gmt`

Leon Simon, *Introduction to Geometric Measure Theory*, Stanford University, 2017/2018 NTU lecture draft; last-update line: 7 March 2018. Author's PDF.

- Normalization: Chapter 1, definitions 2.1–2.2, printed p.11, PDF p.9; equality with Lebesgue outer measure: Theorem 2.6, printed p.12, PDF p.10.
- Area and multiplicity: Chapter 2, Theorem 3.3, printed p.57, PDF p.32; weighted fiber sum: Remark 3.4(2), printed p.58, PDF p.33; curves and submanifolds: Examples 3.5, printed pp.60–61, PDF pp.34–35.
- Coarea: Chapter 2, Theorem 7.3 and weighted Remark 7.4(1), printed p.76, PDF p.42.
- Critical parts of fibers: Theorem 7.5, printed p.80, PDF p.44; the stronger smooth Sard conclusion is distinguished in Remark 7.6.

The PDF has two printed pages per sheet. This version uses the factor `omega_k / 2^k` multiplying sums of diameter powers, where `omega_k` is unit-ball volume. Consequently normalized Hausdorff measure agrees with Lebesgue measure on k-planes. Counting measure is the k=0 convention.

21.4 `hajlasz-geometry`

Piotr Hajłasz, *Geometric Analysis*, University of Pittsburgh, undated author-hosted lecture notes. Author's PDF.

- Normalization: §3.1, printed/PDF p.19; equality with Lebesgue measure: Theorem 3.2, p.20.
- Surface integration: Theorem 3.17 and following discussion, p.25.
- Area and multiplicity with the Lusin N condition: Theorem 3.18 and Remark 3.19, pp.26–27.
- Coarea: §3.4, Theorem 3.23, pp.28–29; radial slicing and projection/Fubini applications, p.29. Almost-everywhere rectifiability: Theorem 3.24, p.29.

Normalization agrees with `simon-gmt`. Two evident dimensional misprints occur in §3.4: the coarea Jacobian should vanish when the derivative has rank less than **m**, not n; the area half of Theorem 3.23 needs outer measure **H^n**, not H^m. The coarea half uses the correct fiber and target dimensions. Our area statements use Simon's 2018 Theorem 3.3 and Remark 3.4(2), whose dimensions are correct.

21.5 `hajlasz-analysis`

Piotr Hajłasz, *Analysis I*, University of Pittsburgh, undated author-hosted lecture notes. Author's PDF.

- Theorem 103 and the Cantor staircase discussion, printed/PDF p.76: a continuous increasing function can have derivative zero almost everywhere while its endpoint increment is positive.
- Absolute continuity and the fundamental theorem: §8.1, Theorem 106, printed/PDF pp.77–78.
- Convolution in L1: Theorem 87, pp.66–67; smooth approximate identities: Theorem 92, pp.70–71.

The Cantor paragraph prints `f'(0) = 0 a.e.`; the variable should be x. The example in this library proves its needed staircase properties directly. Derivative values on its exceptional null set may be assigned arbitrarily for Lebesgue integration, but those assignments do not make the staircase absolutely continuous or Lipschitz.

21.6 `conrad-diff`

Keith Conrad, *Differentiating Under the Integral Sign*, University of Connecticut, undated author-hosted expository notes. Author's PDF.

- A sufficient differentiation theorem: §12, Theorem 12.3, printed/PDF p.16.
- Local parameter bounds illustrating the theorem: Table 2, p.16, and discussion on p.17.

Theorem 12.3 treats continuous integrands and parameter derivatives, dominated by parameter-independent integrable majorants near the parameter in question. Its infinite-interval integrals are stated as improper integrals. The library supplies Lebesgue proofs via dominated convergence of difference quotients; it does not silently equate conditional improper integration with Lebesgue integrability.

21.7 Reading formulas across versions

Use the source IDs above, not a theorem number alone: Simon's 2014 and 2018 drafts number area and coarea differently. The older 2014 course draft, printed p.54, formula 3.4, also has the incorrect target dimension Hˆm; the following weighted formula 3.5 has Hˆn. The library cites the corrected 2018 area statement.

The convention in this directory is scalar integration against positive measures. A parametrization's area Jacobian is nonnegative and counts repeated coverings. An oriented vector flux additionally needs a chosen normal and a signed dot product; it is not determined by the scalar Hausdorff measure alone.

No source access, symbolic calculation, numerical test, or review recorded here is a claim of proof-assistant verification.

22 Verification ledger

The collection contains 16 original worked expositions. Every example received an independent mathematical review by a model agent other than its author. These are recorded mathematical audits, with source and computation checks; **no human peer review or proof-assistant verification is claimed**.

22.1 Independent review of each example

Examples	Author role	Independent reviewer role	Evidence and disposition
E01–E04	Lebesgue examples author	Geometry examples author	Review: all pass; E04's symmetric truncation is explicitly a subset of its open domain.
E05, E06, E14–E16	Parameters/ convergence author	Area/coarea author	Review: all pass; E14's real-measurable assumptions, diagonal subsequence, and source references clarified.
E07–E10	Geometry examples author	Parameters/ convergence author	Review: all pass; no mathematical changes requested.
E11–E13	Area/coarea author	Lebesgue examples author	Review: all pass; E13's empty-interior argument and description of shrinking sets clarified.

The reviewer checked analytic assumptions and derivations, not just final answers. The author signed off the recorded changes. Each review contains SHA256 hashes of the exact reviewed Markdown files; the review manifest and collection checker detect changes after review. No reviewer reviewed their own examples.

The global review separately audits the guide, Hausdorff normalization, theorem dimensions and hypotheses, scalar versus oriented integration, source locators, counterexample atlas, and retrieval metadata. One repaired global statement now restricts sigma-finiteness to the Lebesgue/counting product factors: ambient lower-dimensional Hausdorff measure need not be sigma-finite. The source audit found dimensional misprints in older notes and selected the corrected Simon 2018 area statement.

22.2 Concrete computation checks

Run `python3 integral-examples/checks/check_examples.py` from the repository root. The recorded run in check-results.json performed **92 concrete checks with zero failures**, using SymPy 1.14.0 and mpmath 1.3.0. The JSON records every identity, method, expected value, result, input hash, package version, and computational limitation.

Example	Concrete checks	What those checks support
E01	4	Antiderivative, final value, series coefficient, quadrature in the original order
E02	4	Finite-stage length recurrence/sums and weighted product computation
E03	0	Proof-only: a nonmeasurable set and the null-envelope argument are not computationally verified
E04	8	Primitive identities, iterated values, finite rectangle bound, section quadratures

Example	Concrete checks	What those checks support
E05	3	Exact finite support sums and rectangular exhaustion identities
E06	6	Parameter derivatives, exponential integral, positive/reversed/equal parameter quadratures
E07	5	Speed, weighted length, unweighted length, quadratures
E08	5	Speed/circulation density and once/twice trigonometric integrals
E09	5	Gram determinant, height and area integrals, quadratures
E10	5	Spherical Gram determinant, moment, area, cap formulas
E11	6	Fold determinant, multiplicity-weighted and once-counted integrals
E12	11	Radial Jacobian integrals, power thresholds/formulas, selected quadratures
E13	8	Exact finite-stage Cantor lengths/slopes/masses; no infinite-stage theorem inferred
E14	4	Exact mass and finite tail/convergence-set identities
E15	10	Parameter derivative, antiderivative, symbolic endpoint expression, scaled quadratures
E16	8	Kernel mass, first moment, exponential test-function values, quadratures

The mpmath checks use 70 decimal digits and an absolute-or-relative comparison tolerance of 10^{-50}. This is a numerical consistency test, **not a certified quadrature error bound**. Exact symbolic checks are recorded separately.

Neither symbolic integration nor finite numerical sampling proves measurability, absolute integrability, a limit exchange, a general fiber count, or an infinite-stage Cantor assertion. Those are justified by the written arguments and theorem hypotheses. E03 is explicitly marked proof-only; E13's checks concern finite stages only. No Lean files are supplied or Lean verification asserted for this collection.

22.3 Structural and document checks

- check_collection.py validates all 16 IDs, mandatory sections, source IDs, metadata arrays, local file links, exact review hashes, computation-result hashes, retrieval commands, required deliverables, and the restriction to `integral-examples/` changes.
- metadata.schema.json documents the retrieval schema. The standard-library checker validates the required invariants directly; it does not claim to implement a complete JSON Schema engine.
- build.py generates the retained LaTeX from the Markdown and compiles that exact LaTeX with pdfLaTeX. The build record identifies source/output hashes, tool versions, and any remaining layout warnings.
- collection-check.json records structural verification and final PDF/LaTeX hashes. PDF text and representative pages are also inspected during publication preparation.

22.4 Publication scope and remaining limits

The base and fetched public branch heads are recorded in repository-base.json. The publication branch is `codex/integral-examples-2026-10-15`, based on `e4f337c3aa52ec4c3f2aab078d7c91b32ee0015f`. All additions are under `integral-examples/`. Publication uses an ordinary push to the

new branch, without a pull request, force push, merge, or modifications to other branches. The final task response identifies the actual published commit and verification of remote contents. This source document deliberately does not predict its own future commit hash.

The earlier paper was not supplied or used. The collection is a checked first batch, not an exhaustive atlas of all integration theorems. Its general Lebesgue, area, and coarea theorems remain prerequisites, cited in SOURCES.md. Source PDFs are linked rather than redistributed; online versions can change. Recorded review does not remove the value of checking a theorem's precise hypotheses when adapting an example to a new map, measure, signed weight, or unbounded domain.